Tectonophysics 465 (2009) 60–74

Contents lists available at ScienceDirect

Tectonophysics

journal homepage: www.elsevier.com/locate/tecto

The Yukon–Tanana terrane: Part of North America at ~215 Ma from paleomagnetism of the Taylor Mountain batholith, Alaska

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ARTICLE INFO

Article history:
Received 7 May 2008
Received in revised form 24 September 2008
Accepted 7 October 2008
Available online 18 October 2008

Keywords:
Alaska
Paleomagnetism
Tectonics
Triassic
Yukon-Tanana terrane

ABSTRACT

The massive unmetamorphosed 212 Ma Taylor Mountain batholith underlies ~648 km^2 of the Yukon–Tanana terrane (YTT) in east-central Alaska, and intrudes klippen thrust sheets of the Fortymile River assemblage and the Chicken Metamorphic Complex. Structural evidence indicates that the batholith is neither tilted nor rotated about a vertical axis. Paleomagnetic analysis of 295 specimens from 24 sites in the batholith and cross-cutting mafic dikes isolates a stable primary characteristic remanent magnetization in magnetite with unblocking temperatures of 500 to 585 °C that places the batholith's paleopole at 61.4°N, 102.2°E (N = 17, α_{95} = 5.2°) on the North American apparent polar wander path (APWP) at 215 ± 5 Ma (1σ). The Taylor Mountain paleopole alone and all YTT paleopoles together show that the YTT has been a para-autochthonous part of the North American margin since the Late Triassic. Further, the Intermontane Belt (IMB) paleopoles are distinct from the North American and YTT paleopoles prior to the mid Eocene and provide two APWP segments. One IMB segment includes the Quesnel, Cache Creek, Eastern Coast Belt, and northern Stikine terranes, and the second more-speculative IMB segment includes the central and southwestern Stikine and Methow terranes. This analysis supports tectonic models for the northern Cordillera in which the IMB terranes were obducted onto North America during the Late Cretaceous to mid Eocene, co-eval with the Laramide orogeny.

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1. Introduction

The Yukon–Tanana composite terrane (YTTc) occupies a California-sized tract of the northern Cordillera in Yukon and Alaska (Mortensen, 1992; Foster et al., 1994) (Fig. 1). In the study area of east-central Alaska (Fig. 2), the terrane is composed of "fault-bounded assemblages of ductility-deformed metasedimentary and meta-igneous rocks that are intruded by late- to post-kinematic Late Triassic–Early Jurassic granitoids" and younger intrusives (Dusel-Bacon and Murphy, 2001). The massive Late Triassic to Early Jurassic granitic Taylor Mountain batholith intrudes and stitches together metamorphic rocks of the Fortymile River assemblage, the Nasina assemblage, and the Chicken Metamorphic Complex (Dusel-Bacon et al., 2006, and references therein). From a paleomagnetic perspective, the batholith provides an excellent target for locating the YTTc relative to North America at ~210 Myr ago because the batholith is estimated to have cooled from ~600 °C at 212 Ma to ~300° at 204±9 Ma (Dusel-Bacon et al., 1995, 2002, and references therein), which corresponds to cooling through the unblocking temperature range of magnetite (Pullaiah et al., 1975), potentially enabling the batholith's primary paleopole to be determined. The paleopole, in turn, would permit the testing of the various tectonic hypotheses – autochthonous, para-autochthonous, and allochthonous – that have been proposed for the components of the YTTc since the Late Triassic to Early Jurassic (Foster et al., 1985, Hansen and Dusel-Bacon 1998, Dusel-Bacon et al., 2006).

2. Geology

The Taylor Mountain batholith is located in the YTTc (Coney, 1981) of east-central Alaska, which is part of the Yukon composite terrane of Plafker and Berg (1994). The YTTc is the innermost "suspect" terrane of the northern Cordillera (Monger et al., 1982), abutting against the North American continental margin (Mortensen and Jilson, 1985) (Fig. 1). Many workers have documented the composite nature of the YTTc and proposed different origins and histories for its various components (e.g., Foster et al., 1985, 1994; Hansen and Dusel-Bacon, 1998; Dusel-Bacon et al., 2002, 2006). Tempelman-Kluit (1979) interpreted the YTTc to be a subduction complex that was deformed and thrust northeastward onto the continent in three allochthonous imbricate thrust sheets in the early Mesozoic during arc-continent collision. As such, the YTTc lies between autochthonous or para-autochthonous rocks of North America and demonstrably far-traveled terranes to the southwest (Mortensen, 1992), leading to much debate about its "suspect" origin (e.g. Wheeler et al., 1991; Mortensen, 1992;

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doi:10.1016/j.tecto.2008.10.008

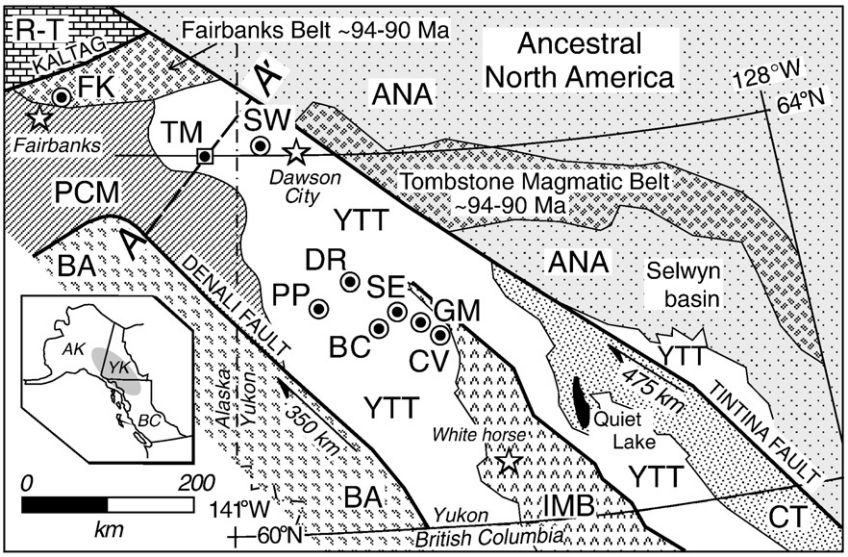


Fig. 1. Location of paleomagnetic collections in the Yukon–Tanana terrane (YTT), adapted from Dusel-Bacon et al. (2006) and Symons et al. (2000). ANA, Ancestral North America; BA, Baja Alaska (northern Insular Belt); BC, Big Creek batholith; CT, Cassiar terrane; CV, Carmacks volcanics (mean location of scattered collection sites); DR, Dawson Range batholith; FK, Fort Knox stock; GM, Granite Mountain batholith; IMB, Intermontane Belt terranes; PCM, para-autochthonous continental margin; PP. Pattison pluton; R–T, Ruby and Tozitna terranes; SE, Seymour Creek stock; SW, Swede Dome stock; TM, Taylor Mountain batholith. References for the collections are given in Table 2. The square surrounding the TM dot is the area shown in Fig. 2.

Foster et al., 1994; Hansen and Dusel-Bacon, 1998). From numerous lines of petrologic, field, and isotopic evidence, the formerly defined "YTTc" is now thought to comprise two packages that experienced different histories between mid-Mississippian and Triassic time: one part rifted away from the continental margin and was outboard of an intervening ocean, and the other stayed behind but was subsequently translated along the margin and overridden by its rifted counterpart as a result of southwest-dipping subduction initiated during the Permian (Dusel-Bacon et al., 2006 and references therein; Nelson et al., 2006). Following Nelson et al. (2006), the rifted portion includes the aforementioned host rocks to the Taylor Mountain batholith and their contiguous equivalents in Yukon is referred to herein as the "Yukon–Tanana terrane" (YTT). The rest of the formerly defined YTTc, which makes up the majority of the terrane in east-central Alaska, is referred to as "parautochthonous continental margin". Recently, Colpron et al. (2006) provided a detailed Devonian to Triassic tectonostratigraphic correlation for the YTT from eastern Alaska to northern British Columbia. They have also provided a broader correlation from Late Proterozoic to Triassic from Yukon to southern British Columbia along with a Devonian to Triassic tectonic model for the Intermontane terranes (Colpron et al., 2007). Also, recent mapping and detrital zircon data from eastern Yukon suggests that detritus from the YTT is first recognized on the western margin of North America by Early Triassic (Smithian) time (Mortensen et al., 2007).

The ovate Taylor Mountain batholith outcrops over an area of ~648 km^2 (Fig. 2), and is composed mostly of medium-grained equigranular granite that grades locally into granodiorite and diorite (Foster et al., 1994). The regional geologic relationships within and around the batholith are well described and illustrated by Hansen and Dusel-Bacon (1998) and Dusel-Bacon et al. (2006), respectively. The

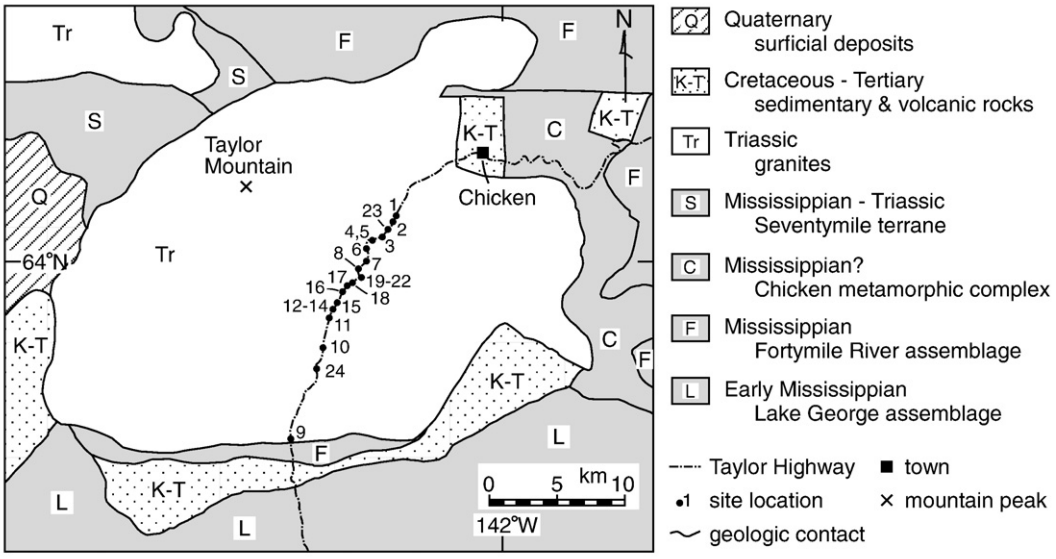


Fig. 2. Location of sample sites in the Taylor Mountain batholith with surrounding geology from Foster (1992) using the revised terminology of Dusel-Bacon et al. (2006).

batholith is generally massive in texture except that it is slightly gneissic along its southern and eastern margins. Mineralogically, the batholith's plagioclase is generally zoned and sericitized, its quartz is strained, and some biotite is altered to chlorite, indicating minor low temperature metamorphism (Foster et al., 1994). The granite has given a U/Pb sphene (titanite) age of 212 Ma (Aleinikoff et al., 1981), 40Ar/39Ar hornblende and biotite ages of 209±3 Ma and 204±9 Ma, respectively (Cushing 1984), and conventional K/Ar hornblende and biotite ages of 195±5 Ma and 198±5 Ma, respectively (Wilson et al., 1985). Importantly, the Taylor Mountain batholith is similar in lithology and age to plutonic rocks of the Klotassin Suite of the Stikine terrane in Yukon, which is also part of the Yukon composite terrane of Plafker and Berg (1994).

The crystalline basement in the YTT of east-central Alaska is composed of metamorphosed and tectonized amphibolite– to lower greenschist-facies rocks (Foster et al., 1994; Hansen and Dusel-Bacon, 1998; Dusel-Bacon et al., 2006). From lowest to highest structural level, the tectonites are divisible into five distinct lithologic assemblages (terminology of Dusel-Bacon et al., 2006), namely: 1) the para-autochthonous, predominantly continental margin strata of the Fairbanks-Chena assemblage; 2) the parautochthonous Lake George assemblage of continental margin metasedimentary rocks and Devono–Mississipian bimodal meta-igneous rocks; 3) the Nasina assemblage of marginal basin metasedimentary rocks, marble, and Mississipian and Permian metatuff; 4) the Fortymile River assemblage and the Chicken Metamorphic Complex of marginal basin metasedimentary rocks and marble, and intermediate and mafic, arc-related meta-igneous rocks of Mississipian age; and, 5) the Seventymile assemblage of oceanic and associated sedimentary strata that are generally not penetratively deformed (Dusel-Bacon et al., 1995; Hansen and Dusel-Bacon, 1998; Dusel-Bacon et al., 2006, and references therein). The assemblages are separated into sheets by flat-lying thrust faults of regional extent, some of which were reactivated by mid-Cretaceous extension (Pavlis et al., 1993; Hansen and Dusel-Bacon, 1998). The massive Late Triassic to Early Jurassic granitoids, including the Taylor Mountain batholith, intrude only rocks of the Fortymile River and Nasina assemblages and the Chicken Metamorphic Complex (Fig. 2). Close to the batholith's eastern contact, rocks of the Chicken Metamorphic Complex have yielded 40Ar/39Ar hornblende pseudo-plateau ages of 213±2 Ma and of 201±5 Ma (Cushing, 1984; Werdon et al., 2001), and 40Ar/39Ar hornblende-biotite from several localities within the Fortymile River assemblage yield ages of about ages 188 to 186 Ma, indicating emplacement of the batholith at ~212 Ma with subsequent uplift and cooling of the Fortymile River assemblage to ~300 °C by ~187 Ma (Hansen et al., 1991; Dusel-Bacon et al., 2001, 2002).

Structurally, Hansen and Dusel-Bacon (1998) concluded that the assemblages in the eastern Yukon–Tanana Upland, which they referred to as the "Upland Tectonites" and are now in sheets separated by near-horizontal ductile faults, were subjected to three deformation events: D1, a subduction-related high pressure and temperature event at, or prior to, 212 Ma that affected only the structurally high packages [the Fortymile River assemblage (their "Taylor Mountain assemblage"), Nasina assemblage (their "Nisutlin assemblage") and the Seventymile assemblage]; D2, a high pressure and temperature collisional event between 212 Ma and 188–185 Ma that resulted in the juxtaposition of the structurally high packages and the underlying continental margin rocks of the Lake George assemblage (their "orthogneiss assemblage"); and, D3, an extensional event between 135 and 109 Ma that resulted in exhumation and cooling of the lower plate Lake George assemblage. Around the southern contacts of the Taylor Mountain batholith at distances up to ~15 km, the D3 event is recognized in the Lake George assemblage metamorphic rocks by five 40Ar/39Ar ages between 119 Ma and 109 Ma in hornblende, biotite and muscovite (Hansen and Dusel-Bacon, 1998; Dusel-Bacon et al., 2002) and by two slightly younger conventional K–Ar ages of ~105 Ma (Wilson et al., 1985).

During the mid-Cretaceous from ~105 to ~85 Ma, massive granitoids of intermediate composition were intruded extensively into the "Upland Tectonites", primarily into the Lake George and

Table 1
Site mean characteristic remanent magnetization directions

Site	Lithology	Specimens measured, averaged	Characteristic remanent magnetization direction				
			Declination, degrees	Inclination, degrees	Radius of 95% Confidence, degrees	Precision parameter *k*	Component
1	Gr	11, 9	309.7	49.4	11.9	19.7	A
2	Gr	11, 11	314.9	60.8	10.3	20.7	A
3	Gr	13, 6	333.0	55.3	7.9	72.4	A
	Gr	13, 7	184.9	86.4	6.4	90.2	B
4	Gr	12, 12	335.7	56.4	7.2	37.4	A
5	Gr	12, 12	308.6	67.9	10.0	19.9	A
6	Gr	13, 9	57.5	83.2	13.3	15.9	B
7	Gr	13, 6	310.8	56.1	11.7	33.6	A
	Gr	13, 6	109.7	88.1	9.2	53.9	B
8	Gr	13, 10	321.7	69.1	10.3	23.0	A
9	Gr	13, 10	131.8	64.1	7.3	44.4	
	Gr	13, 3	354.1	44.1	27.4	21.3	
10	Gr	11, 10	334.9	72.1	6.4	57.4	A
11	Gr	14, 13	316.9	66.4	5.6	55.9	A
12	Md	16, 14	65.8	82.6	3.6	124.7	B
13	Cz	10, 10	34.2	74.7	6.0	66.5	B
14	Gr	12, 11	329.4	70.0	8.1	32.9	A
15	Gr	14, 12	323.9	61.2	9.2	23.2	A
16	Gr	13, 10	322.5	53.8	7.2	46.2	A
17	Gr	11, 11	330.4	60.6	8.1	32.5	A
18	Gr	12, 9	337.3	62.2	8.0	41.9	A
19	Md	13, 13	341.9	52.7	5.7	54.0	
20	Cz	10, 10	344.6	37.1	3.0	266.6	
21	Md	10, 10	300.2	72.8	10.8	21.1	
22	Gr	14, 14	317.5	61.8	8.7	21.9	A
23	Gr	13, 7	332.5	67.7	14.0	19.5	A
	Gr	13, 5	122.9	48.5	13.3	34.0	
24	Gr	11, 9	313.2	73.1	8.5	37.4	A

Notes: lithology: Cz — baked contact zone; Gr — granitoid; and, Md — mafic dike.

Nasina assemblages, and mostly to the south, west and north of the Taylor Mountain batholith (Wilson et al., 1985; Hansen and Dusel-Bacon, 1998). The mid-Cretaceous granitic plutons welded the assemblages of east-central Alaska together and have been related to a northeast-dipping subduction system outboard of, and dipping under, the North American continental margin (Tempelman-Kluit, 1976; Pavlis, 1989; Pavlis et al., 1993; Hansen and Dusel-Bacon, 1998). The closest such granitoid to the batholith is the Walker Fork pluton that crops out ~5 km due east of the batholith (Werdon et al., 2001).

The radiometric age(s) of the mafic dikes cutting the Taylor Mountain batholith is (are) undetermined. Foster et al. (1994) suggested that they could be ~180 Ma late-stage differentiates of the batholith, but that they could be related also to younger more-remote mafic intrusions that range in age from Late Cretacous to Quaternary. Dusel-Bacon and Murphy (2001) noted the presence close to the Taylor Mountain batholith of ~70 Ma alkalic granitoids that give an intraplate trace-element signature (Newberry et al., 1998a) and of subsequent bimodal 60–50 Ma felsic–plutons and mafic dikes that, along with the more-distant Tertiary bimodal volcanics, have an intraplate trace-element signature also (Bacon et al., 1990; Newberry et al., 1998b). Further, Dusel-Bacon and Murphy (2001) showed from prior radiometric evidence and their apatite fission track data that the Taylor Mountain batholith and surrounding host rocks of the Fortymile River assemblage (their "Taylor Mountain" assemblage) cooled from ≥500 to 250 °C by 187 Ma, to 125 °C by 150 Ma, 100 °C by 50 Ma, and to 50 °C by 40 Ma. They attributed the rapid cooling from 100 to 50 Ma to rapid Eocene exhumation due to regional oroclinal rotation of the YTT, dextral strike-slip displacement on the Tintina and Denali faults, and related plate movements.

3. Sampling

The Taylor Mountain batholith is well forested with little outcrop. Fortunately, the Taylor Highway dissects the batholith, providing accessible roadcuts from which 24 sites were collected (Fig. 2) — 19 in the batholith, 3 in cross-cutting mafic dikes, and 2 in the dikes' baked contact zones. At each site, six or more cores were drilled and oriented in situ using a sun compass mostly or magnetic compass if in shadows. From these cores, 10 to 16 standard paleomagnetic specimens of 2.54 cm diameter and 2.20 cm length were prepared, yielding a total of 295 specimens for measurement (Table 1).

4. Experimental methods

The specimens were stored for five years in a magnetically shielded room with an ambient magnetic field intensity of ~0.2% of the Earth's field, allowing their unstable and unwanted viscous remanent magnetization (VRM) components to substantially decay. Then the specimens' residual natural remanent magnetization (NRM) components were measured using an automated vertical-axis 2G Enterprises DC-SQUID magnetometer with a sensitivity of $\sim 2\times10^{-6}$ A/m for a standard specimen.

Initially one specimen per site was alternating field (AF) demagnetized in 12 steps up to 120 mT using a Sapphire Instruments SI-4 demagnetizer, and two additional specimens were thermally (Th) demagnetized in 14 steps up to 630 °C using a Magnetic Measurements (MM-80) demagnetizer. Examining the results of these initial tests, a fourth specimen per site was tested by first AF demagnetizing it at 15 mT and then Th demagnetizing it in 9 further steps from 380 to 600 °C. Excluding 36 remaining specimens that had been glued together and were, therefore, AF demagnetized in 10 steps to 90 mT, the remaining 163 specimens were first AF demagnetized at 15 mT and then Th demagnetized in steps of 400, 450, 500, 530, 550, 570 and 585 or of 380, 420, 460, 500, 520, 540, 560, 575 and 600 °C. Also, saturation isothermal remanent magnetization (SIRM) testing was done on 11 specimens that represented the differing lithologies by first AF demagnetizing them at 130 mT, then pulse magnetizing them in 11 steps up to 900 mT using a Sapphire Instruments SI-6 DC pulse magnetizer, and then AF step demagnetizing them in 9 steps up to 120 mT.

4.1. NRM results

The Taylor Mountain batholith's granitoids yielded a median NRM intensity of $M=9.5\times10^{-2}$ A/m (quartiles, $Q_1=6.0\times10^{-2}$ A/m, $Q_3=13.2\times10^{-2}$ A/m). Notably, site 9 gave two populations of NRM intensities, one of $\sim 9\times10^{-2}$ A/m and the other of $\sim 1\times10^{-3}$ A/m intensity. On AF step demagnetization, most granitoid specimens showed a rapid drop of NRM intensity with a median percentage drop to $M=42\%$ ($Q_1=38\%$, $Q_3=46\%$) by 120 mT except for site 9 that had an anomalously slow drop to only $M=84\%$.

4.2. AF and Th step demagnetization

In an initial test, AF step demagnetization of specimens from each granitic site caused rapid initial remanence intensity decreases but did not isolate their stable characteristic remanent magnetization (ChRM) directions very effectively (Fig. 3). We attributed the ineffectiveness of the AF method to its inability to separate the relatively large modern VRM components and the overprints caused by low-grade burial metamorphism from the primary ChRM components in the granitoids. Conversely, most specimens on the initial Th step demagnetization tests showed a fairly steady NRM intensity drop up to ~450 °C with a corresponding unchanged (Fig. 4 A) or steady shallowing of the NRM inclination into the northwesterly quadrant (Fig. 4 B,C), followed by an intensity plateau to ~525 °C and a sharp intensity drop to ~570 °C with northwesterly intermediate directions, and ending with very weak NRM intensities with erratic directions (Fig. 4). These initial AF and Th step demagnetization results led to the use for the remaining undemagnetized granitic specimens of an initial AF step of 15 mT to suppress the VRM components, followed by Th steps between 380 and 600 °C to isolate the ChRM in the high unblocking temperature range

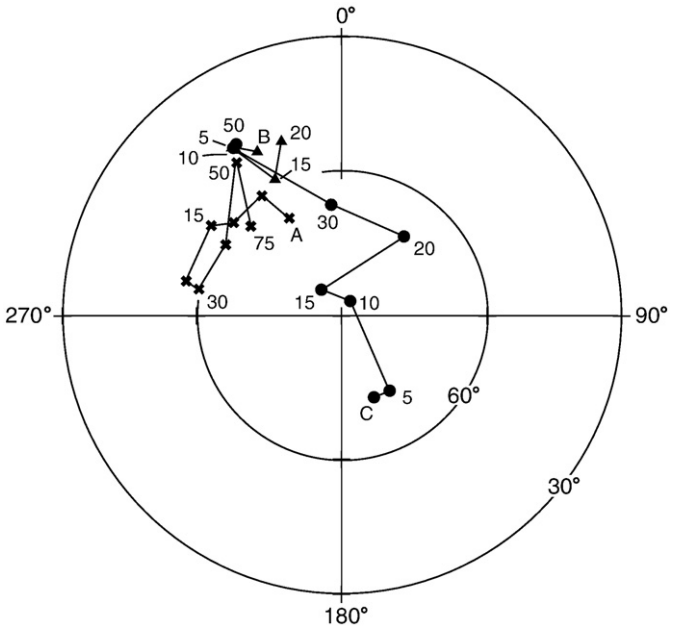


Fig. 3. Interior portion of an equal-area stereonet with the directions for three "good" AF step demagnetized specimens with step intensities indicated in mT. Specimens: A) 110402, B) 180602, and C) 240601 have natural remanent magnetization (NRM) intensities in mA/m×10^{-2} of 7.60, 11.1 and 1.22, respectively. The first two digits of the specimen number designate the sample site.

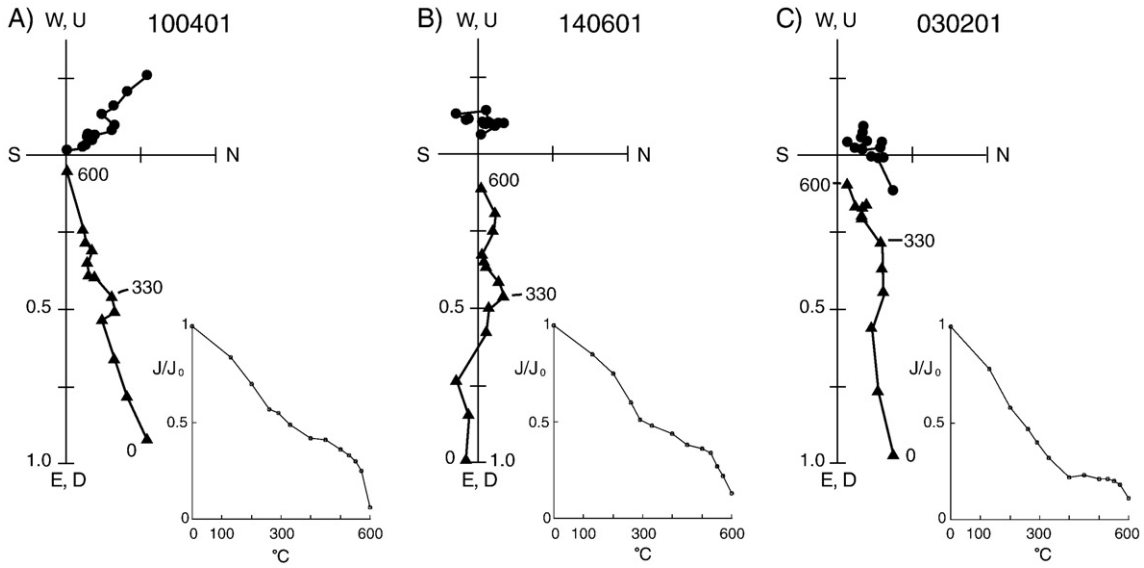


Fig. 4. Vector component plots of the thermal (Th) step demagnetization of example specimens of granite with NRM intensities in mA/m × 10^{-2} of: A) 7.96, B) 11.7, and C) 8.30. Circles [triangles] denote projection on the horizontal [vertical] plane defined by North (N), East (E), South (S) and West (W) [North (N), Down (D), South (S) and Up (U)] axes. Axial values are expressed a ratio of the NRM intensity. Some thermal steps are labeled in °C.

of magnetite and to eliminate the low unblocking temperature overprints from low-grade regional metamorphism, and thereby to isolate the ChRM with an unchanged (Fig. 5 A, B) or shallowing inclination into the northwestern quadrant (Fig. 5 B, C).

4.3. Statistical analysis

The specimens' ChRM directions were determined using vector component plots (Zijderveld, 1967) and least-squares fitting analysis (Kirschvink, 1980). Excluding the AF step demagnetized specimens, the fitting of steps in the ≥500 °C to ≤585 °C temperature range was emphasized, resulting in >94% of the fitted steps being in this range. The site mean ChRM directions were then obtained from the specimen ChRM directions following Fisher (1953) (Table 1). The specimen ChRM directions for 14 of the 19 granitoid sites formed homogenous populations that give site mean ChRM directions to the northwest at an intermediate to moderately steep inclinations, hereafter termed the A component direction. A further three granitoid sites (sites 3, 7 and 23) gave two populations of specimen directions with one population in the A component direction and the other population being discordant. Site 6 gave only a discordant population of specimen directions and site 9 gave two discordant populations. Three of the discordant populations (sites 3a, 6a, and 7a; Table 1) have near-vertical site mean ChRM directions, hereafter termed the B component direction.

The mean A component ChRM direction is at declination D=322.9°, inclination I=62.9° (number of sites N=17, radius of the cone of 95% confidence A_{95}=3.7°, and precision parameter of Fisher

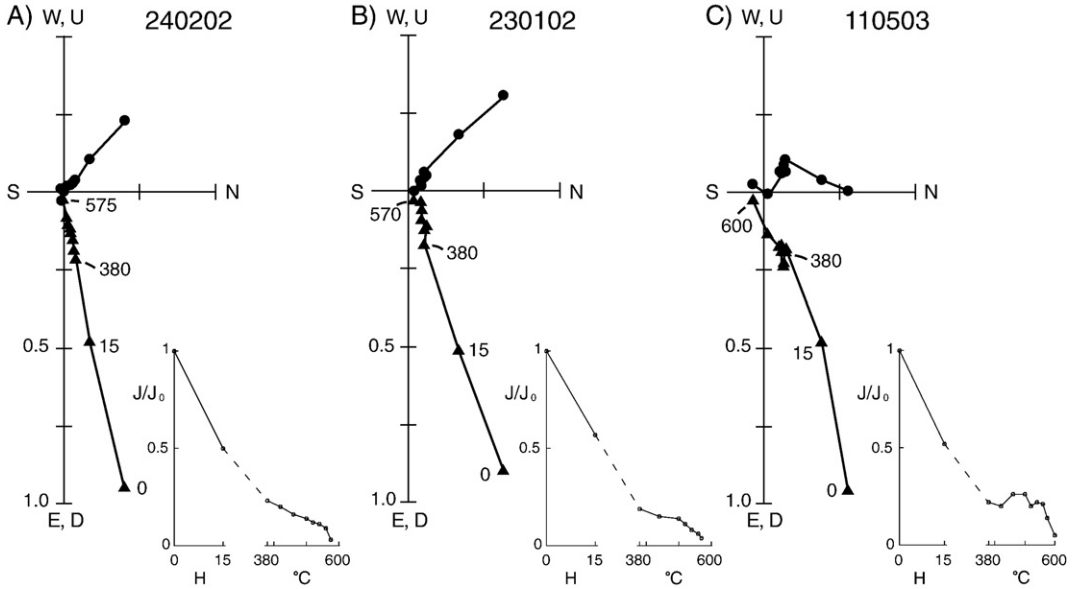


Fig. 5. Vector plots of example granitic specimens on 15 mT alternating field (AF) and then Th step demagnetization with initial NRM intensities in mA/m × 10^{-2} of: A) 6.03, B) 19.3, and C) 10.7.

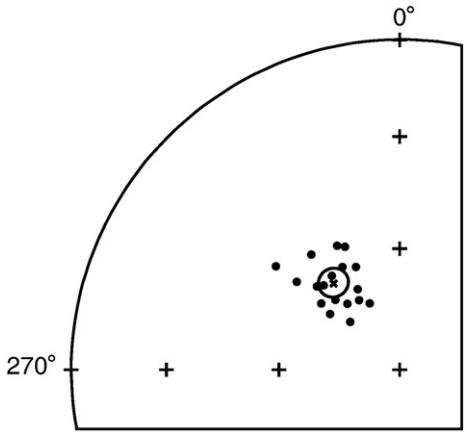


Fig. 6. Site mean directions for the A component in the lower northwest hemisphere of a Schmidt equal-area stereonet with the mean (x) and its surrounding cone of 95% confidence.

(1953) k=96.3) (Fig. 6). In contrast, the B component diverges from the A component by 29° and has a direction of D=94.6°, I=87.6° (N=3, A_{95}=7.3°, k=290).

4.4. Paleomagnetic contact tests

A paleomagnetic baked contact test is positive if a dike and its baked contact zone give a different ChRM direction from remote hostrock (i.e.≥5 times the dike width away), showing that the host rock's ChRM predates the dike's intrusion (Everitt and Clegg, 1962). Conversely, the test is negative if the dike, baked contact zone, and host rock all give the same ChRM direction, indicating either that the hostrock was remagnetized when the dike was emplaced or that both the host rock and dike were subsequently metamorphosed. We attempted three contact tests.

Site 12 sampled a ~1.2 m wide mafic dike that strikes at 94° and dips at 44°S; site 13 was in the dike's granitic baked zone; and, site 14 was in remote granite hostrock from the dike. Both the dike and its contact zone yield stable well-defined B-component ChRM directions that reside in fine-grained magnetite (Fig. 7 A,B) and that are significantly different from the A component direction found at site 14 (Table 1, Fig. 8 A), indicating a positive contact test.

At site 19 a 0.8 m wide mafic dike was sampled that strikes at 45° and dips at 65°SE; site 20 was in the dike's baked contact zone; and, site 22 was in remote hostrock about 9 m away from the dike. Again well-defined ChRM directions, residing in magnetite (Fig. 7 C,D), were isolated in the dike and its baked contact that are located on the northwest side of the A component population, whereas the remote hostrock gives a direction on the A population's southeast side (Fig. 8 B). Thus this contact test is positive but somewhat less convincing because all three directions might be A component directions. Further, a second dike (site 21) of 0.6 m width, strike 50°, and dip 58°SE, was sampled along the same roadcut, and it definitely yielded an A component direction (Fig. 8 B).

4.5. Saturation remanence

When subjected to an applied direct magnetic field, the common magnetic mineral carriers in rocks – goethite, hematite, magnetite, pyrrhotite, and titanomagnetite – acquire magnetizations of differing intensities and stabilities against demagnetization that depend in part on their magnetic domain structure (Dunlop and Özdemir, 1997). The mean SIRM acquisition and decay curves with their root mean square errors are very similar for the eight tested granitoid specimens and the two baked contact specimens, as is the curve for the one dike specimen (Fig. 9). For example, the measured remanence intensity (J) reaches 90% of the saturation intensity (J900) at ~165±20 mT, ~180±25 mT, and ~152 mT, for the granitoids, baked granitoids and dike, respectively (Fig. 9 A). Their

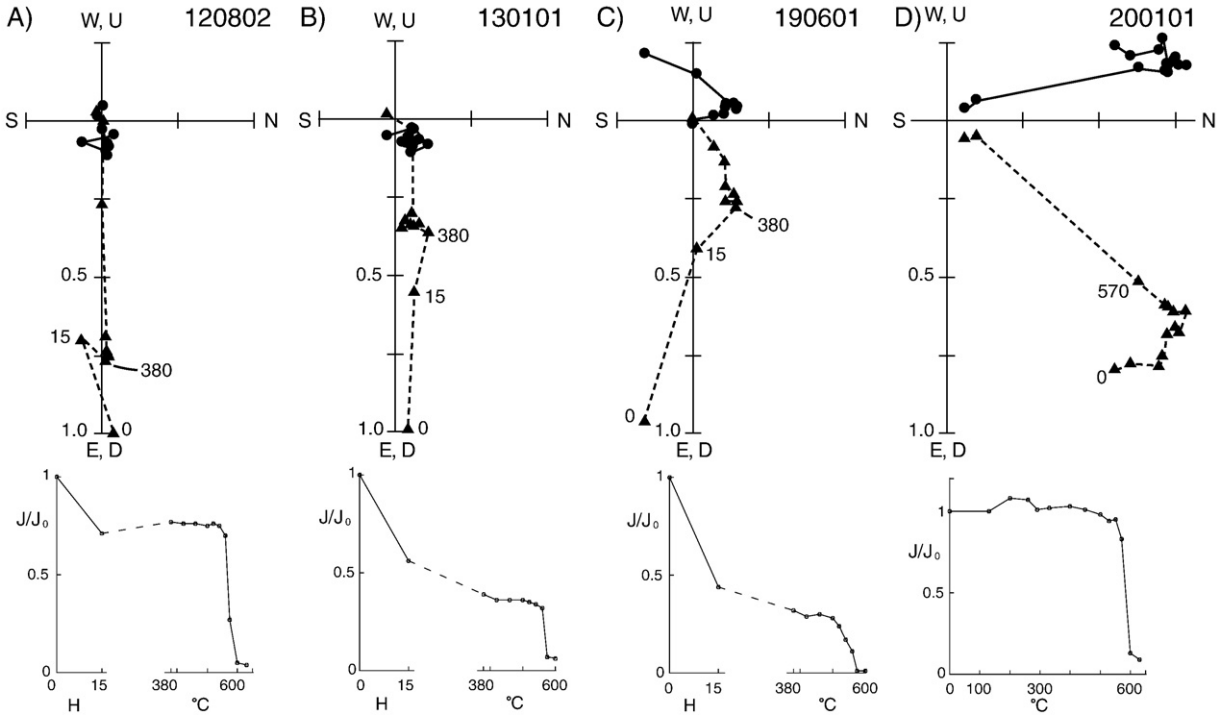


Fig. 7. Vector component plots of AF and then Th step demagnetization of example specimens from mafic dikes (A, C) and their baked contact zones (B, D) with NRM intensities in mA/m×10^{-2} of: A) 2.31, B) 9.60, C) 14.4, and D) 26.4. Other conventions as in Figs. 4 and 5.

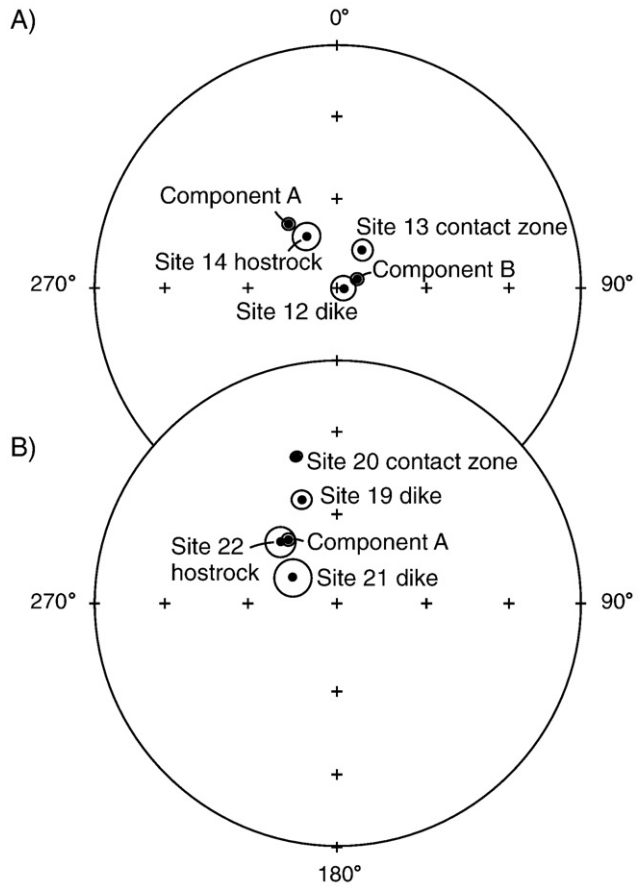


Fig. 8. Equal-area lower hemisphere stereoplots showing site mean directions with their 1σ cones of confidence obtained from baked contact zone tests relative to the A and B component mean directions.

rate of SIRM acquisition, combined with the thermal step demagnetization data, suggests that pseudosingle and single domain magnetite is the major carrier of the ChRM in all tested specimens. This conclusion is supported by the moderately slow rate of decay of SIRM intensity on AF step demagnetization (Fig. 9 B).

5. Discussion

5.1. Apparent polar wander path

The reference apparent polar wander path (APWP) for North America of Besse and Courtillot (2002) is used for dating the ChRM of the Taylor Mountain batholith. The path (Fig. 10) is based on 221 paleopoles from all continents after rotation into the North American frame, runs from 200 Ma to present, and was calculated in 10 Ma increments using a sliding window length of 20 Ma. The path was extended from 200 Ma to 230 Ma as proposed by Tan et al. (2007). The statistical significance of directional populations was determined using the test of McFadden and Lowes (1981). However for the purpose of clarity, all the paleopoles shown in Figs. 10, 11 and 12 are circumscribed by their 1σ cone of 63% confidence rather than the conventional 2σ cone of 95% confidence. This is because two paleopoles, or a paleopole and an APWP, are significantly different at 95% confidence to a close approximation if their 1σ error limits do not overlap.

5.2. A component

The paleopole determined from the A component ChRM direction for the Taylor Mountain batholith is located at 61.4°N, 102.2°E (A_{95} = 5.2). Immediately obvious is that the A paleopole is not significantly different at 95% confidence from either the 210 or 220 Ma reference paleopoles for North America because their 1σ cones of confidence overlap the A paleopole's 1σ cone (Fig. 10). Also obvious is that the best fit of the A paleopole is to the 215 Ma reference paleopole, indicating an age of 215 ± 5 Ma (1σ) for the A component ChRM provided the batholith has not been translated or rotated. Note that the 215 ± 5 Ma (1σ) age for the A component is concordant with the 212 Ma U/Pb sphene (titanite) age of Aleinikoff et al. (1981) for the batholith's emplacement. Further, given the radiometric age dating evidence showing that the Taylor Mountain batholith cooled from ~600 °C at 212 Ma to ~300 °C at 204 ± 9 Ma (Dusel-Bacon et al., 1995), and given the thermal step demagnetization evidence that the ChRM resides mainly in single domain magnetite in a large majority of specimens with laboratory unblocking temperatures of 550 to 585 °C that are equivalent to geologic unblocking temperatures of >500 °C (Pullaiah et al., 1975), it is probable that the measured ChRM formed by ~210 Ma. The important tectonic point to be taken from the concordant radiometric and paleomagnetic ages is that the YTT at

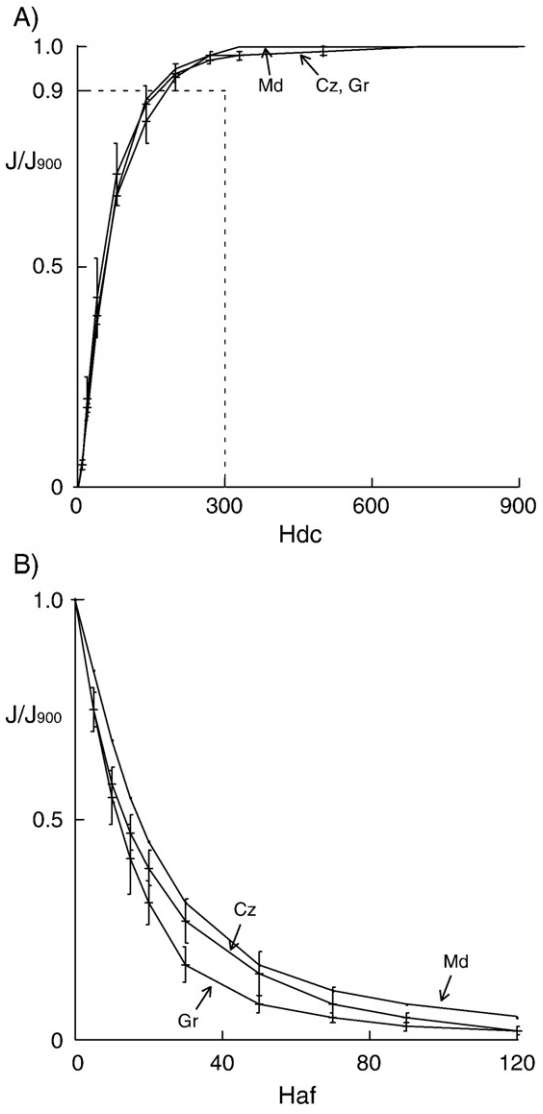


Fig. 9. Saturation isothermal remanent magnetization (SIRM): A) intensity acquisition, and B) demagnetization. J/J900 is the measured intensity (J) as a ratio of the saturation (J900) intensity. The rapid intensity acquisition curves for the 8 granitoid (Gr), 2 baked contact zone (Cz) and one mafic dike (Md) specimens are statistically indistinguishable with all reaching a J/J900 value of 0.90 by 200 mT, indicative of pyrrhotite and/or magnetite.

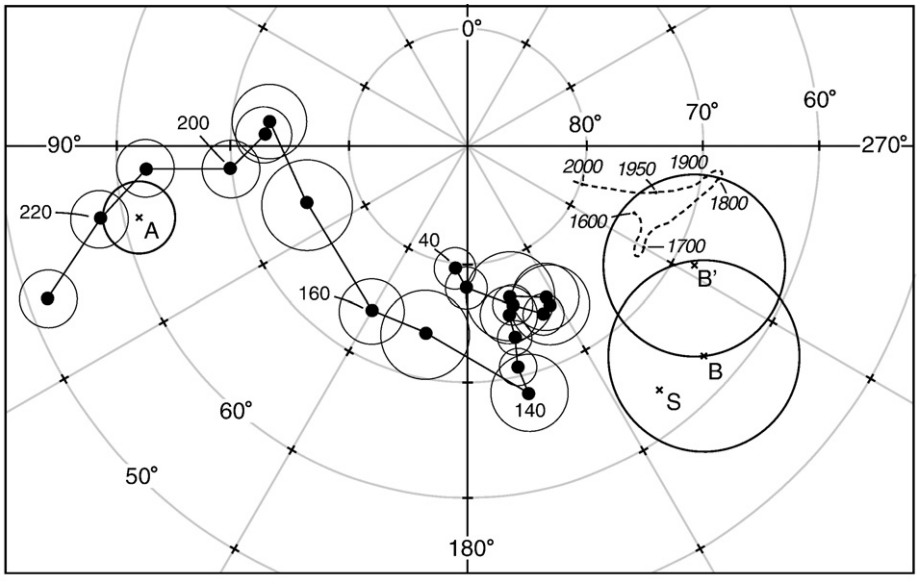


Fig. 10. Apparent polar wander path (APWP) in 10 Myr increments from 0 to 200 Ma of Besse and Courtillot (2002) and on to 230 Ma from Tan et al. (2007) for North America. Also shown are the paleopoles for the A, B and B' characteristic remanent magnetization (ChRM) components obtained for the Taylor Mountain batholith in this study. The paleopoles are circumscribed by their 1σ cones of confidence. S is the location of the Taylor Mountain batholith. The dashed line shows the location of the North Magnetic pole from 1600 to 2000 AD (Geological Survey of Canada, 2005).

Taylor Mountain requires neither translation nor rotation relative to North America to provide an excellent fit of the A paleopole to the APWP, implying that the YTT has been close to its present location as part of the western margin of North America since mid Late Triassic (Norian) time (Gradstein and Ogg, 2004). This observation is confirmed using the method of Demarest (1983) that shows that neither the estimated poleward translation of −2.0°±5.4° nor the estimated clockwise rotation of 4°±7° is statistically significant at 95% confidence (Table 2). Also, this observation fits with the conclusion of Mortensen et al. (2007) from detrital zircon data that the rifted YTT first reached the North American margin by the Early Triassic. Nevertheless, it is important to consider other scenarios for the YTT in which translation or rotation has been suggested.

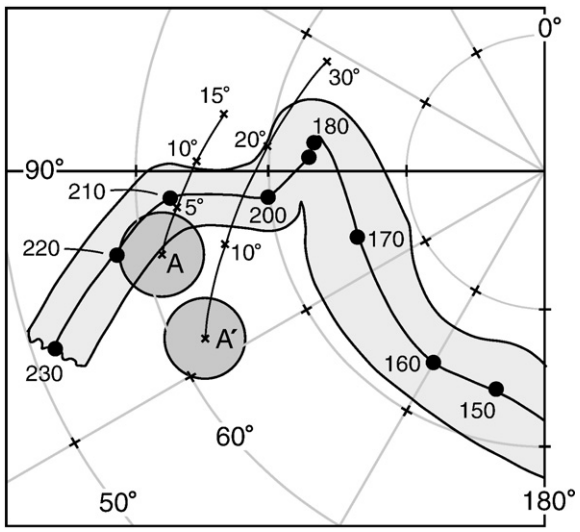


Fig. 11. The ~145 to 230 Ma portion of the North American APWP from Fig. 10 showing the effect on the Taylor Mountain batholith's paleopole of correction for counter-clockwise rotation of the batholith before (A, 0 to 15°) and after (A', 0 to 30°) correction for 475 km of dextral motion on the Tintina fault zone.

5.3. B component

The B component ChRM direction was based initially on the ChRM directions from granitoid sites 3, 6 and 7, giving a paleopole that is well off the APWP for North America and statistically significantly different at 95% confidence from any part of the path, although it is closest to the Early Cretaceous segment of the path. As noted in the discussion of the contact tests, dike site 12 and its baked contact zone site 13 also give B component ChRM directions. Including these two sites in the

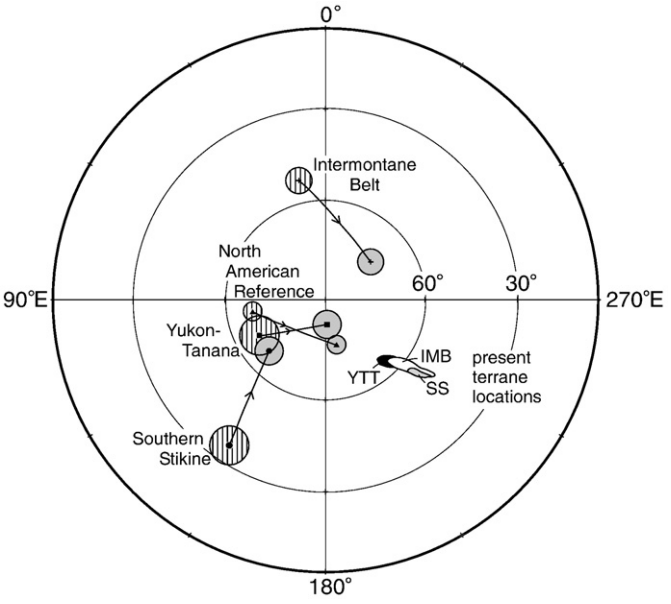


Fig. 12. Equal-area polar projection of the northern hemisphere showing APWP segments from the Late Triassic–Early Jurassic (vertical lined infill) to Late Cretaceous–Paleocene (medium screen infill) for the Intermontane Belt (IMB), Southern Stikine (SS) and Yukon–Tanana (YTT) terranes. Data are listed in Table 4 except that the North American reference segment is from Besse and Courtillot (2002). One sigma error cones are shown for the.North American and IMB paleopoles. Because so few paleopoles are involved for the YTT and SS, the mean one sigma collection values have been averaged for their error cones in this figure.

Table 2
Paleomagnetic estimates of terrane motion for the Yukon–Tanana Terrane

Unit	Name (reference)	Age Ma	Number of sites	Location		Unit paleopole			Reference paleopole			Poleward	Clockwise
				Latitude	Longitude	Latitude	Longitude	α_{95}	Latitude	Longitude	α_{95}	Translation	Rotation
				°N	°E	°N	°E	°	°N	°E	°	°	°
PP	Pattison Pluton (1)	56±1	13	62.4	222.1	86.6	148.1	8.0	76.8	188.9	3.1	8.9 ±6.9	17 ±16
SE	Seymour Creek Stock (2)	69±1	16	62.3	222.8	55.2	202.5	8.4	74.3	205.1	3.5	−0.9±7.3	−94±36
SW	Swede Dome Stock (1)	70±1	17	64.2	219.6	78.6	219.5	5.5	74.3	205.2	3.6	3.2±5.2	20±23
CV	Carmacks Volcanics (3, 4, 5)	70±1	41	62.1	223.7	75.1	129.0	6.3	74.3	205.2	3.6	18.6±5.8	−8±15
FR	Fort Knox Stock (6)*	93±1	18	65.0	212.6	56.5	197.1	5.9	75.8	203.8	5.6	0.2±6.5	−120±35
DR	Dawson Range Batholith (7)	96±4	11	62.6	221.6	80.3	221.7	7.2	76.4	198.2	6.4	1.9±7.7	20±28
GM	Granite Mtn. Batholith (8)	180±10	8	62.3	223.0	74.7	106.9	12.0	73.0	083.4	5.6	−5.1±10.6	−7±18
BC	Big Creek Batholith (9)	180±10	18	62.2	222.6	62.9	146.6	6.5	73.0	083.4	5.6	−9.0±6.9	−38±12
TM	Taylor Mtn. Batholith (10)	212	17	64.0	217.8	61.4	102.2	5.2	61.4	094.9	4.3	−2.0±5.4	−4±7

Numbered references: 1) McCausland et al. (2005); 2) McCausland et al. (2001); 3) Johnston et al. (1996); 4) Wynne et al. (1998); 5) Enkin et al. (2006); 6) Symons et al. (2006), *YTTc; 7) McCausland et al. (2006); 8)McCausland et al. (2002); 9) Symons et al. (2000); 10) this study.

B component average results in a direction of D=56.3°, I=84.6° (N=5, α_{95}=6.8°, k=129.0), giving a paleopole of 68.2°N, 242.4°E (A_{95}=13.3°) that has neither a significantly smaller A_{95} error cone nor a more informative location relative to the APWP to provide a better indication of the B component's age. Noting that the Earth's north magnetic pole has been located for the past ~400 years in the western part of Canada's Arctic Archipelago (Fig. 10) (Geological Survey of Canada, 2005), one possibility is that the B component sites are recording historical lighting strikes (Dunlop and Özdemir, 1997). A second possibility is that the ChRM of the B component is recording scatter from secular variation in the Earth's magnetic field during the Cretaceous–Tertiary intrusion of dikes or extrusion of overlying eroded volcanics.

5.4. Post-100 Ma tectonics of the YTT

In their recent summary Gabrielse et al. (2006) used geological correlations across the Tintina fault zone and adjacent subsidiary faults to constrain dextral displacement to ~60 km between ~100 and ~67 Ma and ~415±15 km between ~67 and ~41 Ma. The ~41 Ma limit was based on an undeformed pluton adjacent to the Tintina fault zone's southeastward extension, the Northern Rocky Mountain Trench fault zone, in central British Columbia. However, seven paleomagnetic studies from Eocene rock units throughout the Canadian Cordillera including Yukon record the end of northwestward displacement of the Intermontane Belt (IMB) terranes relative to North America by ~54 Ma (Symons et al., 2005; McCausland et al., 2006). Together these results imply that ~475 km of dextral displacement on the Tintina fault zone moved the IMB terranes and YTT northwestward between ~100 and ~54 Ma. Correction for the ~475 km (4.3°) of dextral displacement involves sliding the YTT southeasterly between the Tintina and Denali fault zones so that the Taylor Mountain batholith moves from its present location in Alaska to 61.1°N, 226.8°E near Quiet Lake, which is about 80 km northeast of Whitehorse, Yukon (Fig. 1). This results in moving the A paleopole to A' at 62.6°N, 116.3°E, A_{95}=5.3° (Fig. 11). Note that the A' paleopole is now significantly different at 95% confidence from any point on the APWP.

Four tectonic models have been proposed for the YTTc during the ~100 to ~54 Ma interval. From the analysis of the five paleopoles then available, Symons et al. (2005) suggested that the entire YTTc, including both the rifted component and the extended continental margin component (Dusel-Bacon et al., 2006; Nelson et al., 2006), was para-autochthonous by ~100 Ma and simply displaced northwestward on the Tintina fault zone, but that the allochthonous IMB terranes had been thrust over top of the YTT in addition to the Tintina fault zone displacement. They calculated that the IMB underwent 8.3°±7.0° of northwestward translation and 35°±14° of clockwise rotation between ~102 Ma and ~54 Ma, and an additional 16°±6° of clockwise rotation since ~54 Ma.

In the second model, McCausland et al. (2006) concluded from the analysis of paleolatitudinal values from 22 Early Cretaceous to Eocene paleopoles that the YTTc moved with the IMB terranes and was translated northwestward by 7.0°±6.5° (780±720 km) from ~105 to ~54 Ma.

The third model of Enkin et al. (2006) utilized paleolatitudinal data from the 70 Ma Carmacks Group volcanics only. This unit is thought to rest on strata of both the YTT and the Stikine terrane of the IMB, leading Enkin et al. (2006) to propose that both the YTT and IMB moved northwestward by 17.3°±5.5° (1950±600 km) between 68 Ma and the mid Eocene. In the same volume, Gladwin and Johnston (2006) pointed out that, excluding Tintina fault zone displacement, geological evidence from ~104 Ma pinning plutons required that ~1500 km of the displacement registered by the Carmacks volcanics would have to be accommodated by dextral motion on some unknown fault to the northeast of the Selwyn Basin (Fig. 1).

The fourth model, the SAYBIA (Siberia–Alaska–Yukon–British Columbia) model of Johnston (2001), was based on geologic correlations and very selective paleomagnetic data. This model located the YTTc at 27.0°±4.5° (3000±500 km) to the south of its present location at 85 Ma and at 17.3°±6.4° (1900±700 km) to the south at 70 Ma.

Table 3
Paleopoles from the IMB Terranes[a]

Collection (Reference[b])	Terrane[c]	Age (Ma)	Paleopole		
			°N	°E	A_{95}°
ML Mount Lorne plu. (1)	ST	75±3	78.3	290.9	4.3
CC Churn Creek cgl. (2)	QT	89±6	73.2	132.4	4.6
SK Spetch Creek plu. (3)	EC	94±8	73.0	282.0	7.0
SZ Spuzzum plu. (4)	EC	96±2	68.6	349.5	9.8
SQ Silverquick cgl. (5, 6)	MT	97±5	60.2	132.4	5.8
CK Churn Creek volc. (6)	QT	101±4	57.6	299.2	5.7
SF Spius volc. (5, 6)	QT	102±2	63.7	312.4	6.6
MM Mt. McIntyre plu. (7)	ST	109±1	70.5	299.6	5.8
WH Whitehorse bath. (8)	ST	112±4	81.7	285.5	5.5
AX Axelgold intr. (9)	QT	125±5	76.0	327.0	7.5
EB Endako batho. (10, 11)	ST	148±9	58.4	121.8	10.1
FD Fourth July dikes (12)	CC	165±2	59.7	312.0	8.8
FB Fourth July bath. (12)	CC	173±3	65.3	300.2	6.0
TC Teslin Crossing plu. (13)	ST	177±4	84.3	249.4	16.5
HZ Hazelton Gp. volc. (9)	ST	190±5	51.9	130.5	15.8
HV Hazelton Gp. volc. (14)	ST	195±5	24.9	164.0	10.1
GB Guichon bath. (15, 16, 17)	QT	198	49.6	013.5	6.4
CM Copper Mtn. intr. (18, 19)	QT	203	57.3	011.8	3.6
NV Nicola volc. (20)	QT	213±6	52.7	011.3	6.2
TK Takla Gp. volc. (9)	ST	221±3	30.1	140.9	4.5

[a] Table abridged from Table 3 in Symons et al. (2005).
[b] 1, Harris et al., 1999a; 2, Enkin et al., 2003; 3, Vandall 1993; 4, Wynne et al., 1995; 5, Irving et al., 1995; 6, Haskin et al., 2003; 7, Harris et al., 1997a; 8, Harris et al. 1997b; 9, Monger and Irving 1980; 10, Symons 1973b; 11, Symons 1983a; 12, Harris et al., 2003; 13, Harris et al. 1999b; 14, Vandall and Palmer, 1990; 15, Symons 1971; 16, Symons 1983b; 17, Harris et al., 2003; 18, Symons 1973a; 19, Symons and Litalien 1984; 20, Symons, 1984.
[c] CC, Cache Creek; EC, Eastern Coast Belt; MT, Methow; QT, Quesnel; ST, Stikine.

Table 4
Data For Apparent Polar Wander Path (APWP) segments

APWP segment	Average paleopole				
Collection paleopoles	Mean age	Latitude	Longtitude	Precision	Confidence
	Ma	°N	°E	K	A_{95}°
Yukon–Tanana terrane[a]					
– PP, SW, CV, DR	73±14	82.9	180.8	77.1	10.5
– GM, BC, TM	191±16	67.6	119.2	45.5	18.5
Intermontane Belt terranes					
– EP, DF, OL, WP, SP, MV, FC[b]	51±2	84.1	234.0	117.1	5.6
– ML, SK, SZ, CK, SP, MM, WH, AX[c]	102±13	72.3	307.5	59.4	7.2
– FD, FB, TC[c]	172±5	70.6	301.7	30.2	22.9
– GB, CM, NV[c]	205±6	53.2	12.2	423.5	6.0
Southern Stikine terrane[c]					
– CC, SQ	93±4	66.7	132.4	77.8	28.7
– EB[d]	148±9	58.4	121.8	–	10.1
– HZ, HV, TK	202±12	36.4	147.0	16.8	31.0

[a] Data from Table 2.
[b] Data from Table 2 in Symons et al. (2005).
[c] Data from Table 3.
[d] Error estimates for the EB collection only.

Table 2 lists the nine available paleopoles for the YTTc, and gives the translation and rotation values calculated for each. Clearly the 56 Ma Pattison pluton paleopole is not relevant to the translation discussion because its emplacement barely predates the ~54 Ma end of Tintina fault zone displacement (Symons et al., 2005; McCausland et al., 2006). However, the other eight paleopoles in Table 2, including the Fort Knox paleopole in the para-autochthonous continental margin (unit PCM in Fig. 1), do predate 67 Ma that marks the onset of very active Tintina fault displacement (Gabrielse et al., 2006; Enkin et al., 2006). Also, because the North American APWP is generally northwest of the YTT back to 230 Ma (Fig. 10), translation on the Tintina fault zone will be recorded primarily as changes in the poleward translation values. Further, when the ages of the eight remaining paleopoles are examined, they form two groups — one from 69 to 96 Ma (paleopoles SE, SW, CV, FK, and DR) and the other from 180 to 212 Ma (paleopoles GM, BC and TM) (Table 2). When the five 69 to 96 Ma translation values are averaged, they give a mean value of 4.6°±7.1° (510±790 km) at 80±12 Ma in excellent agreement with the known 4.3° (475 km) northwestward displacement of the YTT on the Tintina fault zone. There is, however, a major concern with this translation calculation. It is evident that the 18.6°±5.8° estimate of poleward translation for the Carmacks Group volcanics is significantly different at 95% confidence from the average of the other four 69 to 96 Ma estimates of 1.1°±1.6° (120±175 km), which is less than the geological estimate for Tintina fault displacement.

As will be discussed, the YTT, if accretional, is generally interpreted to have accreted to North America during the Triassic (e.g. Colpron et al., 2006; Mortensen et al., 2007) or Jurassic (e.g. Plafker and Berg, 1994; Hansen and Dusel-Bacon, 1998). The question then is whether the YTT traveled as a constituent part of Baja British Columbia (Baja BC) (Enkin et al., 2006; McCausland et al., 2006) or whether the YTT was overridden on sole thrust faults by Baja BC (Symons et al., 2005). Based on a coherent cluster of 8 of 10 Cretaceous–Paleocene paleopoles from the IMB terranes with an average age of 102±14 Ma (Tables 3 and 4), Symons et al. (2005) obtained a significant northward translation estimate of 8.3°±7.0° (2σ) and a significant clockwise rotation estimate of 51°±14° (2σ) for the IMB. However, all six Cretaceous–Paleocene paleopoles for the YTT (Table 2) are significantly different at 95% confidence from the IMB paleopole cluster, yielding comparable translations but differing rotations ranging from −120° to only +20° clockwise (cw). This contrast in paleopole patterns suggests strongly that the YTT was tectonically independent from the IMB terranes and overridden by them since the mid Early Cretaceous. Further, the distinctively different paleopole patterns for the YTT and IMB over the past ~200 Myr indicates that both the YTT and IMB moved as quasicoherent crustal elements or terranes (Symons et al., 2005). This observation conflicts with the Northern Pacific Rim orogenic Stream (NPRS) or extrusion model proposed by Redfield et al. (2007; see also the comments of Ridgway and Flesch, 2007) for the region in which numerous crustal blocks are variably rotated as they are swept along to give a widely dispersed and incoherent paleopole pattern. Also, the two coherent paleopole patterns are derived from lavas and sedimentary rocks with known dips and from vertical dikes in addition to plutons (Tables 2 and 3). This feature precludes the possibility of invoking a biased regional tilt model to explain the patterns such as that proposed by Butler et al. (1989, 2006) for paleopoles from the Coast Plutonic Complex along the southwestern side of the IMB near Prince Rupert, British Columbia.

5.5. Rotation of the YTT

Another factor to consider is that the Taylor Mountain batholith and its surrounding Chicken Metamorphic Complex and Fortymile River and Nasina assemblages could have been rotated about a vertical axis. The presence of known and postulated near-horizontal thrust faults below the batholith that both separate and subdivide the five hostrock assemblages (Hansen and Dusel-Bacon, 1998) provide potential décollements on which vertical-axis rotations could easily occur during post-emplacement accretion of the thrust sheets. Post-emplacement tilt of the thrust faults is improbable because the klippe that houses both the Nasina assemblage and Taylor Mountain batholith contains an area of ~4000 km^2 or about six times the batholith's size (Dusel-Bacon et al., 2006), and it predates the batholith's intrusion. Such vertical-axis rotations may change the declination of the ChRM but not its inclination, causing the paleopole to move along a small circle with an increasing angle of rotation. For the Taylor Mountain batholith, only a limited range of counterclockwise (ccw) rotations are possible relative to North America because correction for clockwise rotations moves the paleopole to older ages on the APWP. Before correction for Tintina fault zone displacement, the batholith's A paleopole tracks across to progressively lower ChRM ages with increasing rotation angle *ie* on going from 0 to *an* ~12°ccw correction, the A paleopole is not significantly different at 95% confidence from North American reference paleopoles with ages decreasing from ~220 Ma to ~200 Ma (Fig. 11). Similarly, after correction for Tintina fault zone displacement, the A' paleopole on correction for a vertical-axis rotation of ~8 to ~30 ccw is not statistically significantly different at 95% confidence from reference paleopoles with ages from ~210 Ma to ~175 Ma (Fig. 11).

There is evidence for substantial post-Jurassic counterclockwise rotation in Alaska. Following the tectonic model of Plafker and Berg (1994), three events are important. First, the Arctic Alaska terrane, which is across northern Alaska to the north of the Ruby and Tozitna terranes (Fig. 1), underwent an ~66°ccw rotation away from North America from ~120 Ma to ~84 Ma as the Canada Basin opened in the Arctic Ocean (Carey, 1955; Lawver et al., 2002, and references therein). Second, between ~84 Ma and ~66 Ma, much of the dextral displacement occurred on the Tintina fault, causing an ~10°ccw rotation in the Fairbanks Magmatic Belt of the YTT in Alaska relative to its continuation in Yukon, the Tombstone Magmatic Belt (Fig. 1). Third, southwest of the Denali Fault and the YTT are the accreted terranes of Baja Alaska (Packer and Stone, 1974) (Fig. 1) that clearly originated >1000 km south of Alaska (Hillhouse and Coe, 1994). After northward translation and accretion against the YTTc, the Baja Alaska terranes were compressed between Eurasia and North America to form the Alaskan orocline between ~66 Ma and ~50 Ma, which features an ~90°ccw bend that is readily seen in the trace of the Denali fault (Fig. 1) (Plafker and Berg, 1994). An important aspect of all three of these tectonic rotation events is that nearly all of the counterclockwise rotation occurred >100 km to the west of the Taylor Mountain study

area (Plafker and Berg, 1994). Therefore, despite the fact that the ChRM of the 92.5 Ma Fort Knox Stock near Fairbanks, Alaska, which is close to the western end of the YTTc (Fig. 1), records a rotation of 121°±35°ccw (Symons and McCausland, 2006), it is not requisite for the Taylor Mountain batholith's ChRM to also record a substantial counter-clockwise rotation.

Two lines of evidence suggest that the Taylor Mountain batholith and surrounding hostrocks of the YTT have not undergone a vertical-axis rotation of >10°ccw. First, it is possible to draw a tie line through the batholith that is about perpendicular to both the Tintina and Denali fault traces line (line A–A' in Fig. 1). This implies that the fault traces are about parallel and collinear with their southeasterly extensions to 60°N latitude, and that the intervening YTT crust containing the Taylor Mountain region between them has not suffered a significant structural rotation since ~100 Ma when the Tintina fault first became active to move the YTT northwestward between the faults (Gabrielse et al., 2006). Second, as previously noted, the radiometric and thermal step demagnetization data indicate that the batholith acquired its ChRM before ~210 Ma. This observation combined with the ±5 Ma (1σ) error associated with the ChRM's mean direction suggests that the maximum permissible vertical-axis rotation is ~5°±5°ccw for either the A or A' paleopoles.

5.6. Late Triassic–Jurassic YTT tectonics

Tempelman-Kluit (1976), noting their similarity in lithology and age, suggested that the Taylor Mountain batholith and related Late Triassic–Early Jurassic plutons of the YTT and the Klotassin intrusions of the Stikine terrane were the roots of an island arc formed along the margin of the Stikine terrane. Next, Tempelman-Kluit (1979) suggested that the Stikine terrane, with the YTT beneath, rifted away from North America to open the Anvil Ocean. The ocean then closed to form the island arc in Late Triassic, causing the Stikine and YTT terranes to accrete to North America in the Middle Jurassic to earliest Cretaceous by obduction on thrust faults, which trapped the Slide Mountain and Cache Creek terranes in between (Tempelman-Kluit, 1979; Silberling et al., 1992).

The presence of Tethyan fossils of Asian affinity in the Cache Creek terrane amid terranes with fossils of North American affinity (Monger and Ross, 1971) and the advent of plate tectonic theory, led Monger (1977) to suggest that a series of allochthonous crustal fragments or microplates of varied origins accreted sequentially to North America from the southwest to form the Canadian Cordillera (see also Coney et al., 1980; Monger et al., 1982). Foster et al. (1994) considered the YTT subterrane containing the Taylor Mountain batholith to be such a discrete microplate that lay to the north of the Stikine microplate prior to accretion. More recently, English and Johnston (2005) interpreted the Cordilleran terranes in northern British Columbia to have arrived sequentially. They suggested that oceanic seamounts and/or carbonate plateaux with Tethyan fossils were scraped off the top of a subducting seafloor and represented only very minor components that were added to the Cache Creek terrane, thereby mixing temperate North American with Tethyian limestone masses.

The mapping of regional-scale klippen (e.g. Templelman-Kluit, 1979) and of orogen-parallel faults with large displacements (e.g. Gabrielse, 1985), combined with paleomagnetic studies of Mesozoic units suggesting thousands of kilometers of translation in the accreted Cordilleran terranes (e.g. Irving et al., 1996), led to megathrusting models for the Cordillera in which terranes were pushed tens to hundreds of kilometers over top of the North American margin and earlier thrust sheets (e.g. Coney 1989; Hansen 1990; Gehrels et al., 1991a,b). In these models, the YTTc is composed of para-autochthonous continental rocks of the attenuated cratonic margin that were overthrust during the Middle Jurassic by sheets of previously-deformed allochthonous rocks of the Teslin-Taylor Mountain and Nisutlin tectonites (previously used names for the Fortymile River and Nasina assemblages), Quesnel arc, Cache Creek accretionary prism, Seventymile and Slide Mountain oceanic crust, and Stikine arc (Plafker and Berg, 1994;Hansen and Dusel-Bacon 1998; Dusel-Bacon et al., 2006, and references therein). Such megathrusting models have been supported by seismic reflection studies in the Canadian Cordillera (e.g. Cook 1995) that show, for example, the Cache Creek, Quesnel and Slide Mountain terranes resting as klippen on Stikine terrane and North American strata (e.g. Cook et al., 2004; Evenchick et al., 2005). The megathrust model also permits the YTT and IMB paleopoles to be used to give microplate APWPs (Symons et al., 2005) that are far more informative than simple paleolatitudes.

Alternatively, Nelson and Mihalynuk (1993) and Mihalynuk et al. (1994) suggested an oroclinal entrapment model in which the Stikine arc was originally the south-southeast-facing extension of the south-southwest-facing Quesnel arc. They postulated that a series of exotic carbonate plateaux were translated north-northeastward on the Cache Creek seafloor, impacting at the northwest end of the Quesnel terrane in the Late Triassic to create a hinge zone of continental YTT rocks between the two arc segments. Continued north-northeastward seafloor spreading pushed the Quesnel terrane against North America while southeastward seafloor spreading on the north side of the Stikine terrane caused it to under go oroclinal rotation of ~120°, closing first against the Cache Creek plateaux, and continuing around to envelop the plateaux and to close with them against the Quesnel

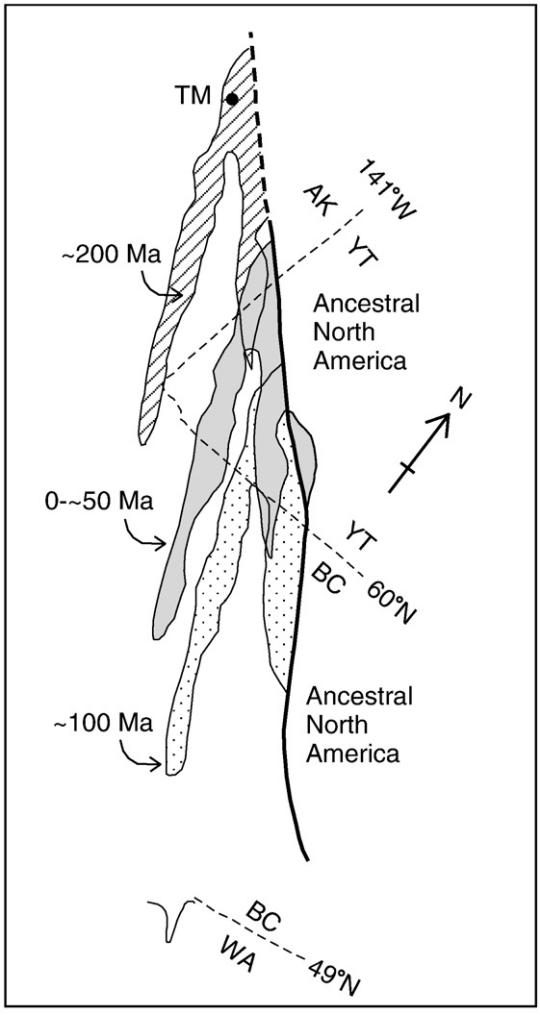


Fig. 13. Schematic showing the location of the presently exposed Yukon–Tanana Terrane (YTT) beside the Tintina and Rocky Mountain Trench fault zones along the western North American margin since the Late Triassic, based on paleomagnetic data. TM is the location of the.Taylor Mountain batholith when emplaced.

terrane in the Middle Jurassic. In this model the YTT is a distinct terrane that is now draped around the northwestern end of the IMB and extends northwestward from it with the Taylor Mountain batholith near its centre (Fig. 1).

In contrast to the preceding tectonic models, Mortensen (1992) argued that Late Triassic–Early Jurassic plutons akin to the Taylor Moutain batholith existed in the Nisling, Stikine, Cache Creek, Quesnel and Slide Mountain terranes as well as the easternmost part of the YTT. Therefore, he deemed them all to be pinning plutons that recorded the Late Triassic–Early Jurassic amalgamation of the "Intermontane Superterrane" with North America.

5.7. APWP segment for the YTT

The three 180 to 212 Ma paleopoles for the YTT (Table 2) yield an average northwestward translation value of −5.4°±2.9°, indicating that they now record southeastward displacement of 590±320 km. Overall, therefore, these three paleopoles indicate that YTT was translated southeastward relative to North America by ~10.0° (1100 km) between ~191±16 Ma and ~80±12 Ma, located from the five translation values, and then northwestward by ~4.6 (510 km) from ~80 Ma to ~54 Ma with negligible rotation relative to North America (Figs. 12 and 13). There are, however, two concerns to note with these translation calculations. First, geobarometry has shown that the 180 Ma Granite Mountain and Big Creek batholiths were emplaced at depths of ~17.5 km and ~16.0 km, respectively (Symons et al., 2000; McCausland et al., 2002), and undoubtedly had large thermal uplift remanence overprints. Incomplete removal of these overprints has the effect of moving the paleopole to lower ages on the APWP and the YTT apparently southeastward. The ChRM in both batholiths is carried by magnetite and demagnetized more rapidly at lower temperatures and alternating fields than, for example, in the Taylor Mountain batholith. Therefore, incomplete removal of these uplift overprints is possible, thereby moving the Granite Mountain and Big Creek paleopoles from ~180 Ma to ~167±2 Ma on the APWP. Thus the location estimate for the YTT at ~191 Ma may be significantly too far north. The second concern is the large time gap of ~110 Myr between the 80±12 Ma and 191±16 Ma groups of YTT paleopoles that needs to be addressed to generate a more definitive APWP segment for the YTT and, thereby, to better assess its tectonic motion.

Despite the above two caveats, and excluding paleopoles from the Fort Knox stock at the YTTc's rotated western end (Symons et al., 2005) and the Seymour Creek stock from next to the Big Creek fault (McCausland et al., 2001), the remaining four Late Cretaceous–Paleocene paleopoles (Table 4) provide an excellent fit to the North American APWP (Fig. 12), and a far better fit than any Paleocene to Late Triassic paleopole in the IMB (Fig. 12). For comparison, the YTT APWP segment has a path length of 21°±5° and an average paleolatitude of 52°±5° that are similar to the coeval North American path values of 28°±3° and 53°±3°, respectively. This result shows clearly that the YTT has been an autochthonous or para-autochthonous part of North America since the start of the Late Triassic, and that the North American APWP can be used as a proxy for the YTT in Yukon. This result fits best with megathrusting models for the northern Cordillera in which both the rifted and the parautochthonous parts of the YTTc originated as parts of the attenuated margin of North America (Coney 1989; Hansen 1990; Gehrels et al., 1991a,b; Plafker and Berg 1994; Hansen and Dusel-Bacon 1998).

5.8. APWP segment for the IMB

The 15 available Eocene-to-present paleopoles for the IMB record a steady progressive clockwise rotation relative to North America, reaching 16°±6° by 54±2 Ma, as well as a nonsignificant 2°±4° northward translation (Symons et al., 2005). Further, 8 of 10 available Cretaceous–Paleocene IMB paleopoles with an average age of 102±14 Ma from the Quesnel, Cache Creek, Stikine, and Eastern Coast Belt terranes (Table 3) form a consistent cluster of ChRM paleopoles that record an additional 35°±14° of clockwise rotation, or 51°±14° in total, plus 8.3°±7.0° (2σ) (915±775 km) of northward translation for the IMB. Symons et al. (2005) interpreted this data to be recording: 1) the initial northward translation and rotation of the IMB on the Kula plate; 2) the rotation of the IMB on collision and obduction onto the North American plate as a thin skin (Fig. 14); and then, 3) the paleolatitudinal translation of the IMB inboard by ~305±125 km since 54±2 Ma (Symons et al., 2003, 2004). In "1)" above, the location of the IMB between ~100 Ma and ~54 Ma was based on the seafloor motion vectors and Kula–Farallon plate geometry provided by Engebretson et al. (1985) and Kelley (1993) for the northeastern Pacific. Acceptance of the alternative Resurrection plate geometry suggested by Haeussler et al. (2003) provides a better explanation both for northwesterly margin-parallel translation of the IMB along the British Columbia–Southeastern Alaska margin of North America from ~70 to ~54 Ma and for the rotation, docking and inland obduction of the IMB over the YTT at ~54 Ma. Moreover, both plate geometries are based on a fixed hotspot reference frame for the Earth's crust, which may well be inaccurate because of mantle flow, meaning that there may be significant errors in tracking translation locations on the seafloor through time (Tarduno, 2007). Nevertheless, using the mid-Cretaceous-to-present rotation and translation estimates for the IMB, Symons et al. (2005) projected the probable location for the IMB's Early Cretaceous–Late Triassic APWP relative to North America. They observed that the available Middle Jurassic paleopoles from the Cache Creek and Stikine terranes in Yukon and northernmost British Columbia and the Early Jurassic–Late Triassic paleopoles from the Quesnel terrane in southern British Columbia fit on the projected APWP, providing a type of oroclinal paleomagnetic fold test. This test supports the interpretation that these three terranes are allochthonous with respect to North America, and that they have traveled together as part of the quasicoherent IMB since the Middle Jurassic at least (Fig. 14).

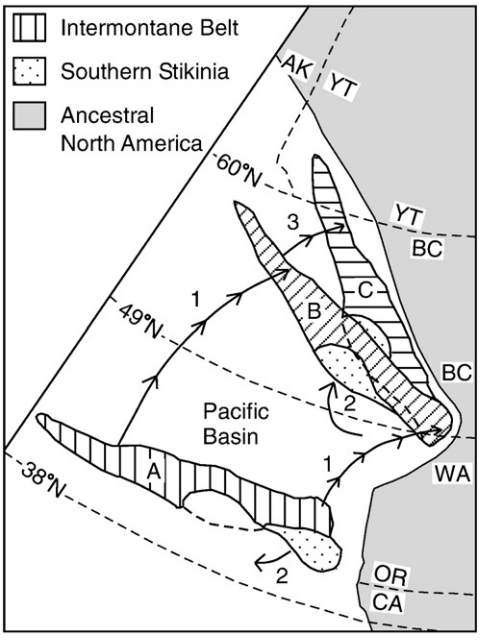


Fig. 14. Schematic, adapted from Symons et al. (2005) showing the rotation of the presently exposed Intermontane Belt (IMB) against ancestral North America, based on paleomagnetic data from ~110 Ma (A) → rotation 1 →~54 Ma (B) → rotation 3 → present (C) to rest on top of the Yukon–Tanana Terrane. The speculative Southern Stikine terrane is suggested to have broken from the southern end of the IMB and Stikinia on initial impact with North America at ~60 Ma and then been rotated by ~180° (rotation 2) to come to rest as a second layer on top the "original" Stikinia.

Accepting the megathrusting model outlined above for the IMB has substantial consequences for Cordilleran tectonic evolution, particularly in terms of timing. For example, it implies that the IMB formed offshore to the south or southwest of its present North American location, and that the tectonic relationships between its several terranes may have no bearing on orogenesis in the Cordillera until the Late Cretaceous. Further, it suggests that the transpressional deformation of the Laramide Orogeny in the Rocky and Mackenzie mountains was caused primarily by the impact and obduction onto North America of the IMB. Also, it implies that some sole faults or décollement zones between the IMB and older autochthonous rocks, such as those imaged by the Lithoprobe seismic surveys across the Yukon (Cook et al., 2004; Evenchick et al., 2005), have been active throughout the Cenozoic in response to subduction of the Kula and Farallon plates beneath North America and the collision of Baja Alaska and the Insular Belt, its southeastward extension (Fig. 1), against North America.

5.9. Southern Stikine APWP segment

A third, more speculative, APWP segment may be present in the accreted Canadian Cordillera that includes paleopoles from the Stikine and Methow terranes in central and southwestern British Columbia (Table 4). This segment lies out in the Pacific Ocean and youngs northward, which is opposite in character to the IMB segment described above (Fig. 12). Mihalynuk et al. (1994) used the ChRM declinations for some of these paleopoles as support for their oroclinal folding hypothesis in which the northwestern extension of the Quesnel terrane is wrapped around the Cache Creek terrane to become the Stikine terrane. However, their model is kinematically difficult to reconcile with the Southern Stikine APWP segment because: 1) the segment requires that rotation postdate 93±4 Ma (Table 4), thus requiring an unknown Late Cretaceous plate with a southeasterly motion vector to be present in the northern Pacific to rotate the 93±4 Ma and older paleopoles; and, 2) the Late Cretaceous Southern Stikine ChRM paleoinclinations are shallower rather than steeper than the coeval IMB paleoinclinations, which is contrary to Mihalynuk et al. (1994) model. A more likely scenario is that the southeastern end of the northeasterly translating Stikine terrane, traveling with the IMB on the Kula plate (Engebretson et al., 1985; Kelley, 1993), impacted first on North America along the Washington–Oregon coastline of the time, was separated from the northwestern end of the Stikine terrane, and was rotated counterclockwise as it overrode North America and/or previously accreted IMB terranes (Fig. 14). Positive features of such a model are that: 1) no unknown oceanic plate is required during the Late Cretaceous to mid Eocene interval; 2) the length of the Late Triassic to Late Cretaceous APWP segment for the Southern Stikine terrane of 32°±5° is similar to the 33°±4° length of the IMB segment, and both are greater that the 28°±3° and 21°±3° lengths of the time equivalent North American and YTT segments; and 3) the average paleolatitude of the Late Triassic to Late Cretaceous APWP segment for the Southern Stikine terrane of 39°±5° is similar to the 41°±4° paleolatitude of the IMB segment, and both are less than the 53°±3° and 52°±5° paleolatitudes for the equivalent North American and YTT segments. Nonetheless, it would be very desirable to increase the paleopole database for the Southern Stikine terrane to confirm this proposed APWP segment and to better define its error limits (Fig. 12).

6. Conclusions

The 212 Ma Taylor Mountain batholith in the YTT of east-central Alaska retains a stable ChRM that resides in magnetite with blocking temperatures in the 500 to 585 °C range. Structural evidence indicates that the batholith has not been significantly tilted or rotated on a vertical axis, but that it has been translated relative to North America by ~475 km on the Tintina fault zone. The batholith's paleopole lies on the North American APWP at ~215±5 Ma, leading to the conclusion that by ~215 Ma the rifted fragment of North America (YTT) rejoined the attenuated continental margin component as a para-autochthonous component of North America to form the Yukon–Tanana composite terrane (YTTc).

When the available paleopoles for the YTT are analyzed as 80±12 Ma and 191±16 Ma paleopole groups, the average paleopoles fall on the North American APWP at close to their expected ages, further suggesting that the YTT is para-autochthons. Also these average paleopoles are significantly different from all co-eval IMB paleopoles, confirming that the YTT is not part of the IMB.

In contrast to the YTT, the IMB terranes were allochthonous with respect to North America prior to ~54±2 Ma with the Late Triassic to Paleocene paleopoles forming two APWP segments. The Quesnel, Cache Creek, northern Stikine and Eastern Coast Belt paleopoles form one segment that is clockwise rotated and northward translated relative to North America. In contrast, the central and southwestern Stikine and Methow terranes form a second more-speculative Southern Stikine segment that is apparently counterclockwise-rotated but also northward translated. The tectonic interrelationships between the YTT, IMB and Southern Stikine APWP segments are attributed to thin-skin tectonics and obduction of the IMB and Southern Stikine terranes over the YTT and North American rocks during the Late Cretaceous to mid Eocene.

Acknowledgements

The authors wish to thank: Craig Hart for pointing out the merits of the Taylor Mountain batholith as a paleomagnetic target; Cynthia Dusel-Bacon, David Scholl and an anonymous reviewer for very helpful referees' reports that have significantly improved this manuscript; the Natural Sciences and Engineering Research Council of Canada for funding this research through Discovery and Lithoprobe grants to D.T.A.S.; and, the Ontario Graduate Fellowship program for fellowship support to P.J.A.M.

References

Aleinikoff, J.N., Dusel-Bacon, C., Foster, H.L., 1981. Geochronologic studies in the Yukon–Tanana Upland, east-central Alaska. In: Albert, N.R.D., Hudson, T. (Eds.), United States Geological Survey in Alaska: Accomplishments during 1979. U.S. Geol. Surv. Circ., 823–B, pp. 34–37.

Bacon, C.R., Foster, H.L., Smith, J.G., 1990. Rhyolitic calderas of the Yukon–Tanana terrane, east-central Alaska: volcanic remnants of a mid-Cretaceous magmatic arc. J. Geophys. Res. 95 (B13), 21451–21461.

Besse, J., Courtillot, V., 2002. Apparent and true polar wander and the geometry of the geomagnetic field over the last 200 Myr. J. Geophys. Res. 107 (B11), 1–31 6.

Butler, R.F., Gehrels, G.E., McClelland, W.C., May, S.R., Klepacki, D., 1989. Discordant paleomagnetic poles from the Canadian Coast Plutonic Complex: regional tilt rather than large-scale displacement? Geology 17, 691–694.

Butler, R.F., Gehrels, G.E., Hart, W., Davidson, C., Crawford, M.L., 2006. Paleomagnetism of Late Jurassic to mid-Cretaceous plutons near Prince Rupert, British Columbia. In: Haggart, J.W., Enkin, R.J., Monger, J.W.H. (Eds.), Paleogeography of the North American Cordillera: Evidence For and Against Large Scale Displacement. Geol. Assoc. Can. Spec. Pap., 46, pp. 171–200.

Carey, S.W., 1955. The orocline concept in geotectonics. Roy. Soc. Tasmania Proc. 89, 255–288.

Colpron, M., Nelson, J.L., Murphy, D.C., 2006. A tectonostratigraphic framework for the pericratonic terranes of the northern Cordillera. In: Colpron, M., Nelson, J.L. (Eds.), Paleozoic Evolution and Metallogeny of Pericratonic Terranes at the Ancient Pacific Margin of North America, Canadian and Alaskan Cordillera. Geol. Assoc. Can. Spec. Pap., 45, pp. 1–23.

Colpron, M., Nelson, J.L., Murphy, D.C., 2007. Northern Cordilleran terranes and their interactions through time. GSA Today 17 (4/5), 4–10.

Coney, P.J., 1981. Accretionary tectonics in western North America. Ariz. Geol. Soc. Dig. 14, 23–37.

Coney, P.J., 1989. Structural aspects of suspect terranes and accretionary tectonics in western North America. J. Struct. Geol. 11, 107–125.

Coney, P.J., Jones, D.L., Monger, J.W.H., 1980. Cordilleran suspect terranes. Nature 288, 329–333.

Cook, F.A., 1995. Lithospheric processes and products in the southern Canadian Cordillera: a Lithoprobe perspective. Can. J. Earth Sci. 34, 1803–1824.

Cook, F.A., Clowes, R.M., Snyder, D.B., van der Velden, A.J., Hall, K.W., Erdmer, P., Evenchick, C.A., 2004. Precambrian crust beneath the Mesozoic northern Canadian

Cordillera discovered by Lithoprobe seismic reflection profiling. Tectonics 23 (TC2010). doi:10.1029/2002TC001412 28 pp.
Cushing, G.W., 1984. The tectonic evolution of the eastern Yukon-Tanana upland, Alaska [Master's thesis]: Albany, State University of New York, 255 pp.
Demarest Jr., H.H., 1983. Error analysis for the determination of tectonic rotation from paleomagnetic data. J. Geophys. Res. 88, 4321–4328.
Dunlop, D.J., Özdemir, Ö., 1997. Rock Magnetism: Fundamentals and Frontiers. Cambridge University Press, Cambridge, U.K. 573 pp.
Dusel-Bacon, C., Hansen, V.L., Scala, J.A., 1995. High pressure amphibolite facies dynamic metamorphism and the Mesozoic tectonic evolution of an ancient continental margin, east-central Alaska. J. Metamorph. Geol. 13, 9–24.
Dusel-Bacon, C., Murphy, J.M., 2001. Apatite fission-track evidence of widespread Eocene heating and exhumation in the Yukon–Tanana Upland, interior Alaska. Can. J. Earth Sci. 38, 1191–1204.
Dusel-Bacon, C., Lanphere, M.A., Sharp, W.D., Layer, P.W., Hanson, V.L., 2002. Mesozoic thermal history and timing of structural events for the Yukon–Tanana Upland, east-central Alaska — 40Ar/39Ar data from metamorphic and plutonic rocks. Can. J. Earth Sci. 39, 1013–1051.
Dusel-Bacon, C., Hopkins, M.J., Mortensen, J.K., Dashevsky, S.S., Bressler, J.R., Day, W.C., 2006. Paleozoic tectonic and metallogenic evolution of the pericratonic rocks of east-central Alaska and adjacent Yukon Territory. In: Colpron, M., Nelson, J.L. (Eds.), Paleozoic evolution and metallogeny of pericratonic terranes at the ancient Pacific margin of North America, Canadian and Alaskan Cordillera. Geol. Assoc. Can. Spec. Pap., 45, pp. 25–74.
Engebretson, D.C., Cox, A., Gordon, R.G., 1985. Relative motions between oceanic and continental plates in the Pacific Basin. Geol. Soc. Am. Spec. Pap. 206 59 pp.
English, J.M., Johnston, S.T., 2005. Collisional orogenesis in the northern Canadian Cordillera: Implications for Cordilleran crustal structure, ophiolite emplacement, continental growth, and the terrane hypothesis. Earth Planet. Sci. Lett. 232, 333–344.
Enkin, R.J., Baker, J., Mahoney, J.B., Riesterer, J., 2003. Deciphering shallow paleomagnetic inclinations: 2. Implications from Late Cretaceous strata overlapping the Insular/Intermontane Superterrane boundary in the southern Canadian Cordillera. J. Geophys. Res. 108, 1–19.
Enkin, R.J., Johnston, S.T., Larson, K.P., Baker, J., 2006. Paleomagnetism of the 70 Ma Carmacks Group at Solitary Mountain, Yukon, confirms and extends controversial results: further evidence for the Baja British Columbia model. In: Haggart, J.W., Enkin, R.J., Monger, J.W.H. (Eds.), Paleogeography of the North American Cordillera: Evidence For and Against Large-Scale Displacements. Geol. Assoc. Can., pp. 221–232.
Evenchick, C.A., Gabrielse, H., Snyder, D., 2005. Crustal structure and lithology of the northern Canadian Cordillera: alternative interpretations of SNORCLE seismic reflection lines 2a and 2b. Can. J. Earth Sci. 42, 1149–1161.
Everitt, C.W.F., Clegg, J.A., 1962. A field test of palaeomagnetic stability. Geophys. J. R. Astron. Soc 6, 312–319.
Fisher, R.A., 1953. Dispersion on a sphere. Proc. Roy. Soc. London A217, pp. 295–305.
Foster, H.L., 1992. Geologic map of the eastern Yukon-Tanana Region, Alaska. U.S. Geol. Surv., Open-file Rept 92-313, scale 1:500000.
Foster, H.L., Cushing, G.W., Keith, T.E.C., Laird, J., 1985. Early Mesozoic tectonic history of the Boundary area, east-central Alaska. Geophys. Res. Lett. 12, 553–556.
Foster, H.L., Keith, T.E.C., Menzie, W.D., 1994. Geology of the Yukon–Tanana area of east-central Alaska. In: Plafker, G., Berg, H.C. (Eds.), The Geology of North America, G-1, The Geology of Alaska. Geol. Soc. Am., pp. 205–240.
Gabrielse, H., 1985. Major dextral transcurrent displacements along the northern Rocky Mountain Trench and related lineaments in north-central British Columbia. Geol. Soc. Amer. Bull. 96, 1–16.
Gabrielse, H., Murphy, D.C., Mortensen, J.K., 2006. Cretaceous and Cenozoic dextral orogen-parallel displacements, magmatism, and paleogeography, north-central Canadian Cordillera. In: Haggart, J.W., Enkin, R.J., Monger, J.W.H. (Eds.), Paleogeography of the North American Cordillera: Evidence For and Against Large-Scale Displacements. Geol. Assoc. Can., pp. 255–276.
Gehrels, G.E., McClelland, W.C., Samson, S.D., Patchett, P.J., Brew, D.A., 1991a. U–Pb geochronology of Late Cretaceous and early Tertiary plutons in the northern Coast Mountains batholith. Can. J. Earth Sci. 28, 899–911.
Gehrels, G.E., McClelland, W.C., Samson, S.D., Patchett, P.J., 1991b. U–Pb geochronology of detrital zircons from a continental margin assemblage in the northern Coast Mountains, southeastern Alaska. Can. J. Earth Sci. 28, 1285–1300.
Geological Survey of Canada, 2005, Geomagnetism: long term movement of the north magnetic pole in depth (http://gsc.nrcan.gc.ca/geomag/nmp/long_mvt_nmp2_e.php).
Gladwin, K., Johnston, S.T., 2006. Mid-Cretaceous pinning of accreted terranes to miogeoclinal assemblages in the northern Cordillera: irreconcilable with paleomagnetic data? In: Haggart, J.W., Enkin, R.J., Monger, J.W.H. (Eds.), Paleogeography of the North American Cordillera: Evidence For and Against Large-Scale Displacements. Geol. Assoc. Can., pp. 299–306.
Gradstein, F.M., Ogg, A.G., Smith, A.G., (and 33 others), 2004. A Geologic Time Scale 2004. Cambridge University Press. 464 pp.
Haeussler, P.J., Bradley, D.C., Wells, R.E., Miller, M.L., 2003. Life and death of the Resurrection plate: evidence for its existence and subduction in the northeastern Pacific in Paleocene–Eocene time. Geol. Soc. Amer. Bull. 115, 867–880.
Hansen, V., 1990. Yukon–Tanana terrane: a partial acquittal. Geology 18, 365–369.
Hansen, V.L., Dusel-Bacon, C., 1998. Structural and kinematic evolution of the Yukon–Tanana upland tectonites, east-central Alaska: a record of late Paleozoic to Mesozoic crustal assembly. Geol. Soc. Amer. Bull. 110, 211–230.
Hansen, V.L., Heizler, M.T., Harrison, T.M., 1991. Mesozoic thermal evolution of the Yukon–Tanana composite terrane: new evidence from 40Ar/39Ar data. Tectonics 10, 51–76.
Harris, M.J., Symons, D.T.A., Blackburn, W.H., Hart, C.J.R., 1997a. Paleomagnetic study of the mid-Cretaceous Mount McIntyre pluton, Whitehorse map area (105 D), southern Yukon Territory. In: Roots, C.F. (Ed.), Yukon Exploration and Geology 1996. Explor. & Geol. Services Div., Yukon, Indian and Northern Affairs Canada, Whitehorse, Y.T., pp. 122–130.
Harris, M.J., Symons, D.T.A., Blackburn, W.H., Hart, C.J.R., 1997b. Paleomagnetic and geobarometric study of the mid-Cretaceous Whitehorse Pluton, Yukon Territory. Can. J. Earth Sci. 34, 1379–1391.
Harris, M.J., Symons, D.T.A., Blackburn, W.H., Hart, C.J.R., 1999a. Paleomagnetic and geobarometric study of the Late Cretaceous Mount Lorne stock, southern Yukon Territory. Can. J. Earth Sci 36, 905–915.
Harris, M.J., Symons, D.T.A., Hart, C.J.R., Blackburn, W.H., 1999b. Jurassic plate motions of the Yukon: a paleomagnetic study of the Teslin Crossing stock (NTS 105 E/7). In: Roots, C.F., Emond, D.S. (Eds.), Yukon Exploration and Geology 1998. Explor. & Geol. Services Div. Yukon, Indian and Northern Affairs Canada, Whitehorse, Y.T., pp. 155–166.
Harris, M.J., Symons, D.T.A., Blackburn, W.H., Villeneuve, M.J., Hart, C.J.R., 2003. Travels of the Cache Creek Terrane: a paleomagnetic, geobarometric and 40Ar/39Ar study of the Jurassic Fourth of July batholith, northernmost British Columbia, Canada. Tectonophysics 326, 137–159.
Haskin, M.L., Enkin, R.J., Mahoney, J.B., Mustard, P.S., Baker, J., 2003. Deciphering shallow paleomagnetic inclinations: 1. Implications from correlation of Albian volcanic rocks along the Insular/Intermontane Superterrane boundary in the southern Canadian Cordillera. J. Geophys. Res. 108, 1–17.
Hillhouse, J.W., Coe, R.S., 1994. Paleomagnetic data from Alaska. In: Plafker, G., Berg, H. C. (Eds.), The Geology of North America, G-1, The Geology of Alaska. Geol. Soc. Am., pp. 797–812.
Irving, E., Thorkelson, D.J., Wheadon, P.M., Enkin, R.J., 1995. Paleomagnetism of the Spences Bridge Group and northward displacement of the Intermontane Belt, British Columbia: a second look. J. Geophys. Res. 100, 6057–6071.
Irving, E., Wynne, P.J., Thorkelson, D.J., Schiarizza, P., 1996. Large (1000 to 4000 km) northward movements of tectonic domains in the northern Cordillera, 83 to 45 Ma. J. Geophys. Res. 101, 17901–17916.
Johnston, S.T., 2001. The great Alaskan terrane wreck: reconciliation of paleomagnetic and geologic data in the northern Cordillera. Earth Planet. Sci. Lett. 193, 259–272.
Johnston, S.T., Wynne, P.J., Francis, D., Hart, C.J.R., Enkin, R.J., Engebretson, D.C., 1996. Yellowstone in Yukon: the Late Cretaceous Carmacks Group. Geology 24, 997–1000.
Kelley, K., 1993. Relative motions between North America and oceanic plates of the Pacific basin during the past 130 million years. M.Sc. Western Washington Univ., Bellingham, WA, U.S.A. 89 pp.
Kirschvink, J.L., 1980. The least squares line and plane and the analysis of paleomagnetic data. Geophys. J. R. Astron. Soc 62, 699–718.
Lawver, L.A., Grantz, A., Gahagan, L.M., 2002. Plate kinematic evolution of the present Arctic region since the Ordovician. In: Miller, E.L., Grantz, A., Klemperer, S.L. (Eds.), Tectonic evolution of the Bering Shelf-Chukchi Sea-Arctic Margin and Adjacent Landmasses. Boulder, Colorado, Geol. Soc. Am. Spec. Paper 360, pp. 333–358.
McCausland, P.J.A., Symons, D.T.A., Hart, C.J.R., Blackburn, W.H., 2001. Paleomagnetic study of the Late Cretaceous Seymour Creek stock, Yukon: minimal geotectonic motion of the Yukon–Tanana Terrane. In: Emond, D.S., Weston, L.W. (Eds.), Yukon Exploration and Geology 2000, pp. 207–216.
McCausland, P.J.A., Symons, D.T.A., Hart, C.J.R., Blackburn, W.H., 2002. Paleomagnetism and geobarometry of the Granite Mountain batholith, Yukon: minimal geotectonic motion of the Yukon–Tanana terrane relative to North America. In: Emond, D.S., Weston, L.H., Lewis, L.L. (Eds.), Yukon Exploration and Geology 2001. Explor. & Geol. Services Div., Yukon. Indian and Northern Affairs Canada, Whitehorse, pp. 163–177.
McCausland, P.J.A., Symons, D.T.A., Hart, C.J.R., 2005. Rethinking "Yellowstone in Yukon" and Baja British Columbia: paleomagnetism of the Late Cretaceous Swede Dome stock, northern Canadian Cordillera. J. Geophys. Res. 110. doi:10.1029/2005JB003742 13 pp.
McCausland, P.J.A., Symons, D.T.A., Hart, C.J.R., Blackburn, W.H., 2006. Assembly of the northern Cordillera: new paleomagnetic evidence for coherent, moderate Jurassic to Eocene motion of the Intermontane Belt and Yukon–Tanana terranes. In: Haggart, J. W., Enkin, R.J., Monger, J.W.H. (Eds.), Paleogeography of the North American Cordillera: Evidence For and Against Large-Scale Displacements. Geol. Assoc. Can., pp. 147–170.
McFadden, P.L., Lowes, F.J., 1981. The discrimination of mean directions drawn from Fisher distributions. Geophys. J. R. Astron. Soc. 67, 19–33.
Mihalynuk, M.G., Nelson, J.A., Diakow, L.J., 1994. Cache Creek terrane entrapment: Oroclinal paradox within the Canadian Cordillera. Tectonics 13, 575–595.
Monger, J.W.H., 1977. Upper Paleozoic rocks of the western Canadian Cordillera and their bearing on Cordilleran evolution. Can. J. Earth Sci. 14, 1832–1859.
Monger, J.W.H., Ross, C.A., 1971. Distribution of fusulinaceans in the western Canadian Cordillera. Can. J. Earth Sci. 8, 259–278.
Monger, J.W.H., Irving, E., 1980. Northward displacement of north-central British Columbia. Nature (London) 285, 289–294.
Monger, J.W.H., Price, R.A., Tempelman-Kluit, D.J., 1982. Tectonic accretion and the origin of the two major metamorphic and plutonic welts in the Canadian Cordillera. Geology 10, 70–75.
Mortensen, J.K., 1992. Pre–Mid-Mesozoic tectonic evolution of the Yukon–Tanana Terrane, Yukon and Alaska. Tectonics 11, 836–853.
Mortensen, J.K., Jilson, G.A., 1985. Evolution of the Yukon–Tanana terrane: evidence from southeastern Yukon Territory. Geology 13, 806–810.
Mortensen, J.K., Beranek, L., Murphy, D.C., 2007. Permo-Triassic orogeny in the northern Cordillera: Sonoma north? Geol. Soc. Am. Abstracts with Programs 39 (4), 69.
Nelson, J., Mihalynuk, M., 1993. Cache Creek ocean: closure or enclosure? Geology 21, 173–176.
Nelson, J.L., Colpron, M., Piercey, S.J., Dusel-Bacon, C., Murphy, D.C., Roots, C.F., 2006. Paleozoic tectonic and metallogenetic evolution of pericratonic terranes in Yukon, northern British Columbia and eastern Alaska. In: Colpron, M., Nelson, J.L. (Eds.),

Paleozoic evolution and metallogeny of pericratonic terranes at the ancient Pacific margin of North America, Canadian and Alaskan Cordillera. Geol. Assoc. Can. Spec. Pap., 45, pp. 323–360.
Newberry, R.J., Bundtzen, T.K., Mortensen, J.K., Weber, F.R., 1998a. Petrology, geochemistry, age, and significance of two foliated intrusions in the Fairbanks District, Alaska. In: Riehle, J.R., Gray, J.E. (Eds.), Geologic studies in Alaska by the U.S. Geological Survey, 1996. U.S. Geol. Surv., Prof. Paper 1595, pp. 117–129.
Newberry, R.J., Layer, P.W., Burleigh, R.E., Solie, D.N., 1998b. New 40Ar/39Ar dates for intrusions and mineral prospects in the eastern Yukon–Tanana Terrane, Alaska. In: Riehle, J.R., Gray, J.E. (Eds.), Geologic studies in Alaska by the U.S. Geological Survey, 1996. U.S. Geol. Surv., Prof. Paper 1595, pp. 131–160.
Packer, D.R., Stone, D.B., 1974. Paleomagnetism of Jurassic rocks from southern Alaska and the tectonic implications. Can. J. Earth Sci. 11, 976–997.
Pavlis, T.L., 1989. Middle Cretaceous orogenesis in the northern Cordillera: a Mediterranean analog of collision-related extensional tectonics. Geology 17, 947–950.
Pavlis, T.L., Sisson, V.B., Foster, H.L., Nokleberg, W.J., Plafker, G., 1993. Mid Cretaceous extensional tectonics of the Yukon–Tanana terrane, Trans-Alaska Crustal Transect (TACT), east-central Alaska. Tectonics 12, 103–122.
Plafker, G., Berg, H.C., 1994. Overview of the geology and tectonic evolution of Alaska. In: Plafker, G., Berg, H.C. (Eds.), The Geology of North America, G-1, The Geology of Alaska. Geol. Soc. Am., pp. 989–1021.
Pullaiah, G., Irving, E., Buchan, K.L., Dunlop, D.J., 1975. Magnetization changes caused by burial and uplift. Earth Planet. Sci. Lett. 28, 133–143.
Redfield, T.F., Scholl, D.W., Fitzgerald, P.G., Beck, M.E., 2007. Escape tectonics and the extrusion of Alaska: Past, present, and future. Geology 35, 1039–1042.
Ridgway, K.D., Flesch, L.M., 2007. Cenozoic tectonic processes along the southern Alaska convergent margin. Geology 35, 1055–1056.
Silberling, N.J., Jones, D.L., Monger, J.W.H., Coney, P.J., 1992. Lithotectonic terrane map of the North American Cordillera. U.S. Geol. Surv. Map I-2176, scale 1:5000000.
Symons, D.T.A., 1971. Paleomagnetism of the Triassic Guichon batholith and rotation in the Interior Plateau, British Columbia. Can. J. Earth Sci. 8, 1388–1396.
Symons, D.T.A., 1973a. Unit correlations and tectonic rotation from paleomagnetism of the Triassic Copper Mountain intrusions, B.C. Geol. Surv. Can. 73–19, 11–28.
Symons, D.T.A., 1973b. Paleomagnetic results from the Jurassic Topley intrusions near Endako, British Columbia. Can. J. Earth Sci. 10, 1099–1108.
Symons, D.T.A., 1983a. Further paleomagnetic results from the Jurassic Topley Intrusions in the Stikina subterrane of British Columbia. Geophys. Res. Lett. 10, 1065–1068.
Symons, D.T.A., 1983b. New paleomagnetic data for the Triassic Guichon batholith of south-central British Columbia and their bearing on Terrane I tectonics. Can. J. Earth Sci. 20, 1340–1344.
Symons, D.T.A., 1984. Paleomagnetism of the Triassic Nicola volcanics and geotectonics of the Quesnellia subterrane of Terrane I, British Columbia. Geodynamics 2, 229–244.
Symons, D.T.A., Litalien, C.R., 1984. Paleomagnetism of the Lower Jurassic Copper Mountain intrusions and the geotectonics of Terrane I, British Columbia. Geophys. Res. Lett. 11, 685–688.
Symons, D.T.A., McCausland, P.J.A., 2006. Paleomagnetism of the Fort Knox Stock, Alaska, and rotation of the Yukon–Tanana terrane after 92.5 Ma. Tectonophysics 419, 13–26.
Symons, D.T.A., Williams, P.R., McCausland, P.J.A., Harris, M.J., Hart, C.J.R., Blackburn, W.H., 2000. Paleomagnetism and geobarometry of the Big Creek Batholith suggests that Yukon–Tanana Terrane has been a parautochthon since Early Jurassic. Tectonophysics 326, 57–72.
Symons, D.T.A., Erdmer, P., McCausland, P.J.A., 2003. New 42 Ma cratonic North American paleomagnetic pole from the Yukon underscores another Cordilleran paleomagnetism-geology conundrum. Can. J. Earth Sci. 40, 1321–1334.
Symons, D.T.A., Hart, C.J.R., Harris, M.J., McCausland, P.J.A., 2004. Paleomagnetism of the Eocene Flat Creek pluton, Yukon: Tectonics of the Intermontane terranes and Mackenzie Mountains. Geol. Soc. Am. Spec. Pap. 383, 187–204.
Symons, D.T.A., Harris, M.J., McCausland, P.J.A., Blackburn, W.H., Hart, C.J.R., 2005. Mesozoic–Cenozoic Paleomagnetism of the Intermontane and Yukon–Tanana Terranes, Canadian Cordillera. Can. J. Earth Sci. 42, 1163–1185.
Tan, X., Kodama, K.P., Gilder, S., Courtillot, V., 2007. Rock magnetic evidence for inclination shallowing in the Passaic Formation red beds from the Newark Basin and a systematic bias of the Late Triassic apparent polar wander path for North America. Earth Planet. Sci. Lett. 254, 345–357.
Tarduno, J.A., 2007. On the motion of Hawaii and other mantle plumes. Chem. Geol. 241, 234–237.
Tempelman-Kluit, D.J., 1976. The Yukon crystalline terrane: enigma in the Canadian Cordillera. Geol. Soc. Amer. Bull. 87, 1343–1357.
Tempelman-Kluit, D.J., 1979. Transported Cataclasite, Ophiolite, and Granodiorite in Yukon: evidence of Arc-continent Collision. Geol. Surv. Can. 79–14 27 pp.
Vandall, T.A., Palmer, H.C., 1990. Canadian Cordilleran displacement: palaeomagnetic results from the Early Jurassic Hazelton Group, Terrane I, British Columbia, Canada. Geophys. J. Internat. 103, 609–619.
Vandall, T.A., 1993. Cretaceous Coast Belt paleomagnetic data from the Spetch Creek pluton, British Columbia: evidence for the “tilt and moderate displacement” model. Can. J. Earth Sci. 30, 1037–1048.
Werdon, M.B., Newberry, R.J., Szumigala, D.J., 2001. Bedrock geologic map of the Eagle A-2 quadrangle, Fortymile mining district, Alaska. Alaska Div. Geol. & Geophys. Surv. Prelim. Interpretive Rep. 2001-3b, scale: 1:63,360.
Wheeler, J.O., Brookfield, A.J., Gabrielse, H., Monger, J.W.H., Tipper, H.W., Woodsworth, G.J., 1991. Terrane map of the Canadian Cordillera. Geol. Surv. Can. Map 1713A, scale 1:2000000.
Wilson, F.H., Smith, J.G., Shew, N., 1985. Review of radiometric data from the Yukon crystalline terrane, Alaska and Yukon Territory. Can. J. Earth Sci. 22, 525–537.
Wynne, P.J., Irving, E., Maxson, J.A., Kleinspehn, K.J., 1995. Paleomagnetism of the Upper Cretaceous strata of Mount Tatlow: evidence for 3000 km of northward displacement of the eastern Coast Belt, British Columbia. J. Geophys. Res. 100, 6073–6091.
Wynne, P.J., Enkin, R.J., Baker, J., Johnston, S.T., Hart, C.J.R., 1998. The big flush: paleomagnetic signature of a 70 Ma regional hydrothermal event in displaced rocks of the northern Canadian Cordillera. Can. J. Earth Sci. 35, 657–671.
Zijderveld, J.D.A., 1967. A.C. demagnetization of rocks: analysis of results. In: Creer, K.M.C., Collinson, D.W., Runcorn, S.K. (Eds.), Methods in Paleomagnetism: Elsevier. Amsterdam, The Netherlands, pp. 254–286.